Class 9
Chemistry
Solutions

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1. Solutions

Worksheet – 1

Level – I

1. A	4. B	7. D	10. B
2. B	5. A	8. B	
3. C	6. D	9. A	

Level – II

1. C	2. C	3. A	4. D	5.B	6. C	7. A	8. A
9. B	10. B	11. A	12. A	13. A	14. C	15. C	16. B
17. A,C,D	18. A	19. C	20. A	21. B	22. a – p, b – q, c – r, d –s		
23. B,C	24.A,B,C	25. B	26. D	27. C	28. a – s, b – r, c – q, d – p		
29. A,B							

4. $\frac{V}{V}\% = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$

Volume of Solute = 50ml

Volume of Solution = Volume of Solute + Volume of Solvent

$= 50 + 500 = 550$ml

$\frac{\text{V}}{\text{V}}\% = \frac{50}{550} \times 100 = \frac{500}{55} = 9.09$

4th Option

5. $M = \frac{n_{\text{solute}}}{\text{Vol of Solution (ml)}} \times 1000$

$M = \frac{0.006}{100} \times 1000$

M = 0.06M

2nd Option

6. Density of solution $= 0.96 \frac{\text{g}}{\text{ml}}$

1ml → 0.96gr

100ml → 0.96×100

mass of solution = 96 g

mass of solute = 15 g

mass % $= \frac{\text{mass of solute}}{\text{mass of solution}} \times 100$

$= \frac{15}{96} \times 100$

$= 15.625\%$

3rd Option

7. $\text{F} = \frac{\text{Wsolute}}{\text{Gmw solute}} \times \frac{100}{V_{(m1)}}$

$\text{F} = \frac{47.5}{95} \times \frac{1000}{500}$

$F = 1F$

1st Option

8. 10^6ml of H_2O has 155 g Urea.

9. $mass\ \% = \frac{\text{mass of solute}}{\text{mass of solute + mass of solvent}} \times 100$

$10 = \frac{X}{(300+X)} \times 100$

$3000 + 10X = X100$

$3000 = 100x - 10x$

$3000 = 90x$

$X = \frac{3000}{90} = 33.33$

2nd Option

10. $M = \frac{n_{solute}}{V_{(Lit)}}$

$0.5 = \frac{n}{2}$

n = 1

2nd Option

11. 3.5 Grams of NaCl in 100 g of water

H_2O water Present in seawater = 100 − 3.5

= 96.5 g (or.

= 0.0965Kg

1st Option

12. $M = \frac{\text{No.of milli moles of Solution}}{\text{Volumein ml}}$

Vol.of solution $= \frac{10}{0.08}$

= 125ml

1st Option

13. 1st Option

35ppm CO means, 35 g of CO is present in 10^6 g of air

∴ Volume of air $= \frac{\text{mass of air}}{\text{density of air}}$

$= \frac{10^6}{1.3} = 7.69 \times 10^5$ L

7.69×10^5 Lof air has → 35 g of CO

∴ 1 of air has = x ?

$= \frac{35}{7.69 \times 10^5} = 4.55 \times 10^{-5}$ gCO

14. $w = \frac{M \times gmw \times V_{ml}}{1000}$

$W = \frac{0.1 \times 278 \times 100}{1000}$

$W = 2.78g$

15. 3rd option

$\frac{10}{180}$ moles are present in 100 cc i.e 0.1 L

1 mole of glucose will present in $= \frac{100 \times 180}{10} = 1800$ml

16. $N_1 = 0.1N$HCl

$V_1 = 100$ml of HCl

$N_2 = 0.1\text{M} \times 2 = 0.2\text{N}H_2So_4$

$V_2 = 50\text{ml of } H_2So_4$

$$N_{res} = \frac{N_1V_1 + N_2V_2}{V_1 + V_2}$$

$$= \frac{100 \times 0.1 + 50 \times 0.2}{150}$$

$$N_{\underset{(NaOH)}{res}} = \underset{(\text{ res Acid })}{0.133}$$

$N_1V_1 = N_2V_2$

$100 \times 0.2 = 0.133 \times 150$

$20 = 20$ Equal

2nd Option

17. 1,3,4

$$w/w\% = \frac{10}{110} \times 100$$

$= 9.09$

1st Option

19. $W\%\alpha \frac{1}{\text{w.ofsolution}}$

$$\frac{(w\%)_1}{(w\%)_2} = \frac{wof\ I^{nd}\ \text{Solution}}{\text{wof I } I^{st}\ \text{solution}}$$

$$\frac{6}{30} = \frac{200}{X}$$

$$X = \frac{30 \times 200}{6} = 1000$$

Amount of $H_2O = 1000 - 200 = 800$ g

20. $\text{v/v\%} = \frac{\text{Vol . of solute}}{\text{Vol.of solute + Vol of Solvent}} \times 100$

$$20 = \frac{70}{(70 + X)} \times 100$$

$1400 + 20x = 70 \times 100$

$1400 + 20x = 70 \times 100$

$x = \frac{5600}{20} = 280\text{ml}$

21. $\text{w/w\%} = \frac{W_{\text{solute}}}{W_{\text{solute}} + W_{\text{solvent}}} \times 100$

$$= \frac{15}{15 + 60} \times 100 = \frac{15}{75} \times 100$$

$= 20\%$ 2nd Option

24. 1,2,3

$$M = \frac{W}{GMW} \times \frac{1000}{Vm_1}$$

$$M = \frac{5.85}{58.5} \times \frac{1000^{10}}{100}$$

$M = 1M$

$$M = \frac{111}{111} \times \frac{1000}{1000}$$

$M = 1M$

$$M = \frac{0.32}{32} \times \frac{1000}{10}$$

$M = 1M$

25. $M = \frac{d \times W\% \times 10}{\text{mol.wt}}$

$$M = \frac{15 \times 10 \times 1.10}{98} = 1.68$$

2nd Option

26. $M = \frac{98 \times 10 \times 1.95}{98} = 19.5$

4th Option

27. $M = \frac{98 \times 10 \times 1.84}{98}$

$M = 18.4$

3rd Option

Worksheet – 2
Level – I

1.D 2.C 3. D 4.B 5.B 6.A 7.B 8.C
9.C 10.A

3. Number of moles = 0.1
wt. of solvent (w.= 100 g = 0.1 kg
$m = \frac{n}{Wkg} = \frac{0.1}{0.1} = 1$ m

4. Number of moles of solute = $n_1 = 6$ moles
Number of moles of solution = $n_1 + n_2 \Rightarrow 100$ moles
Mole fraction solute $(x_1) = \frac{n_1}{n_1+n_2}$
$= \frac{6}{100} = 0.06$

5. Wt. of Urea = $w_1 = 60$ g
GMW of Urea = $M_1 = 60$ g
Wt. of Urea = $W_2 = 90$ g
GMW of Urea = $M_2 = 18$ g
No. of moles of Urea = $n_1 = \frac{W_1}{M_1}$
$= \frac{60}{60} \Rightarrow n_1 = 1$
no.of moles of water = $n_2 = \frac{90}{18}$
$n_2 = 5$
Mole fraction of solute $(x_1) = \frac{1}{1+5} = \frac{1}{6}$

8. molarity = 3 m
Vol. of water (V.= 1000ml
wt. of water = W = 1000 g = 1 kg
$(\therefore d = 1$ g/ml)
$m = \frac{n}{W_{kg}} \Rightarrow n = m \times w$
$m = 3 \times 1 = 3$ moles
option 3

9. wt.of NaOH = $W = 4$ g
Gmw of NaOH = 40 g
Wt.ofwater $W = 1$ kg

$$m = \frac{W}{Gmw} \times \frac{1}{W_{kg}} = \frac{4}{40} \times \frac{1}{1}$$

$$m = 0.1m$$

10. $N = \frac{W}{4EW} \times \frac{1000}{V_{ml}}$

$$N = \frac{35}{56} \times \frac{1000}{400}$$

$$N = 1.56N$$

1st Option

Level – II

1. B	2. C	3. A	4. C	5.D	6. D	7. C	8. A
9. A	10. A	11.C	12.B	13.C	14. B	15.A	16. C
17. C	18. B,D	19.B	20.B	21.A	22. B		

23. a – p, b – q, c – r, d – p,s 24. 2.68 25. B,C 26. A 27.a – p, b – q, c – r, d –t
28. A 29. a – t; b – p; c – q;d – r 30. a – s; b – p; c – t; d

1. N of $H_2SO_4 = 0.5 \times 2$

$\Rightarrow 1N$

$N_1 = 1$

$V_1 = 1$ Lit

$N_2 = ?$

$V_2 = 10$ Lit

$N_1V_1 = N_2V_2$

$1 \times 1 = x.10$

$x = \frac{1}{10} \Rightarrow 0.1N$

2nd option

2. Milli equivalents $= N \times V_{ml}$

$= 500 \times 0.2 = 100$

3rd Option

3. N of $H_3PO_3 = 0.1 \times 2 = 0.2$ N

$(H_3PO_3 \rightarrow 2H^+ + HPO_3^{-2})$

1st Option

4. $\overset{+3}{M_2}O_3 \rightarrow \underset{M\uparrow}{\overset{+5}{O_5}}$

Change Oxidation No = 2

Total change = 4

Equivalent weight $= \frac{m \cdot wt}{4}$

4th option

5. $n_{Nacl} = \frac{5.85}{58.5} = 0.1$

$n_{H_2O} = \frac{90}{18} = 5$

$X_{Nacl} = \frac{0.1}{5.1} \Rightarrow \frac{1}{51}$

$= 0.0196$

4th option

6. Equivalent $wt = \frac{Gmw}{\text{Change of cation / Anio}}$

$= \frac{\text{Molecular weight}}{4}$

4th **option**

7. $\underset{100}{M}O_2 \rightarrow \underset{124}{MO}$

$W_0 = 124 - 100 = 24$

$Eq_M = \frac{W_M}{W_0} \times 8 = \frac{100}{24} \times 8$

$= 33.33$

3rd Option

8. $N = \frac{W}{GEW} \times \frac{1}{v_{\text{Lit}}}$

$= \frac{31.5}{63} \times \frac{1}{2.5} = 0.2$ N

1st Option

9. $\text{N} = \frac{W}{GEW} \times \frac{1000}{V_{ml}}$

$\frac{1}{10} = \frac{W}{100} \times \frac{1000}{100}$

$W = 1gr$

1st Option

10. $n_{\text{Nacl}} = 1$

$n_{H_2O} = \frac{1000}{18} = 55.55$

$X_{\text{Nacl}} = \frac{1}{55.55 + 1} X_{\text{Nacl}} = \frac{1}{56.55} = 0.0177$

1st Option

11. $\text{N} = \frac{W}{GEW} \times \frac{1000}{V_{ml}}$

$N = \frac{10}{49} \times \frac{1000}{100} = \frac{100}{49}$

$N = 2.04N$

3rd Option

12. $N_a V_a = N_b V_b$

$0.2 \times V_a = 0.5 \times 1$

$V_1 = \frac{0.5}{0.2} \Rightarrow 2.5$

2nd Option

13. $n_{H_2O_2} = \frac{20}{34} = 0.588$

$n_{H_2O_2} = \frac{80}{18} = 4.44$

$X_{H_2O} = \frac{4.44}{4.44 + 0.588}$

$= 0.883$

3rd Option

14.

$\underbrace{MnO_4^- - - - - \rightarrow M_n}_{+5 \text{ (change)}}{}^{+2}$

$N = 5 \times M$

$\therefore 5M = N$

$HCl \rightarrow \underset{M=N}{1H^+} + Cl^-$

$M = N$

15. $M = \frac{W}{GMW} \times \frac{1000}{W_{kg}}$

$= \frac{3}{30} \times \frac{1000}{250}$ $M = 0.4m$

1st Option

16. $X_{H_2O} + X_{CH_3OH} = 1$

$X_{H_2O} = 0.4$

$$X_{H_2O} = \frac{n_{H_2O}}{n_{H_2O} + n_{CH_3OH}}$$

$0.4 = \frac{\frac{w}{18}}{1.5 + \frac{w}{18}}$

$0.4\left(1.5 + \frac{w}{18}\right) = \frac{w}{18}$

$0.6 + \frac{0.4 \times w}{18} = \frac{w}{18}$

$\frac{10.8 + 0.4 \times w}{18} = \frac{w}{18}$

$10.8 = 0.6w$

$w = \frac{10.8}{0.6} = 18$

3rd Option

(or)

$0.6 = \frac{1.5}{1.5 + \frac{w}{18}}$

$0.9 + \frac{0.6 \times w}{18} = 1.5$

$\frac{16.2 + 0.6w}{18} = 1.5$

$16.2 + 0.6w = 27$

$0.6W = 10.8$

$W = 18$

17. $\left(\frac{N \times V_{ml}}{1000}\right)_b = (GEN)_9$

$\frac{20 \times 0.1}{1000} = \frac{wt}{GEN}$

$\frac{2}{1000} = \frac{wt}{63}$

$wt = \frac{63 \times 2}{1000}$

$= 0.126$

Option -3

19. Option = 2

$HNO_3 \rightarrow H^{\oplus} + NO_3^-$

M = N

20.

(i)

$M = \frac{w}{Gmw} \times \frac{1000}{V_{mL}}$

$\frac{8}{56} \times \frac{1000}{100}$

= 1.4 N

(ii)

$N = \frac{W\% \times 10}{GEW}$

$= \frac{\frac{6}{106} \times 100 \times 10}{40}$

= 1.41 N

(iii) 1 N

(iv)

$N = \frac{24.5}{49} \times \frac{1}{1}$ Option -2

= 0.5 N

21. $\frac{N \times V_{ml}}{1000} = GEW$

$\frac{0.1 \times 1000}{1000} = \frac{0.63}{GEW}$

GEW = 63

Option -1

22. $GEW = N \times V_{Lit}$

$= 0.1 \times 0.25$

$= 0.05$

Option - 2

24. Volume strength =

$5.6 \times N$

$15 = 5.6 \times N$

$N = \frac{15}{5.6} = 2.68$

27. $a - q, s: b - p, c - s; d - t$

(a) $N = \frac{20}{40} \times \frac{1000}{250} = 1$ N

(b) $N = \frac{10}{40} \times \frac{1000}{250} = 2$ N

(c) $N = \frac{30}{40} \times \frac{1000}{250} = 3$ N

(d) $N = \frac{40}{40} \times \frac{1000}{250} = 4$ N

Worksheet – 3

Level – I

1. C	4. C	7. A	10. B
2. D	5. A	8. C	11. C
3. A	6. C	9. C	

Solub ility $= \frac{\text{Mass of solute}}{\text{Mass of Solvent}} \times 100$

2.

$= \frac{2.5}{25} \times 100$

Level – II

1. B	2. A	3. A	4. B	5. C	6. C	7. B
8. A	9. B	10. A	11. D	12. A	13. B	14. D
15. B	16. D	17. A,B,C	18. A,C,D	19. A	20. A	21. A
22. a – s, b – r, c – q d – p		23. A,C	24. A,B,C,D	25. A,B,C,D	26. A	27. D
28. C	29. a – r,b – p, c – q d–s		30. A,B,C			

1. Option -2

M_2 = molarityUrea = 0.2 m

Wt. of glu cose =? (w_1)

Wt.of. water = W_1 = 500 g

Gmw of glu cose = 180 g

$m_1 = m_2$

$\frac{w_1}{GmW} \times \frac{1000}{W_1} = m_2$

$\frac{W_1}{180} \times \frac{1000}{500} = 0.2$

W_1 = 18 g

2. Option-1

Acc to Raoult'slaw

R.L.V.P = molefraction of solute

$\frac{\Delta_p}{P^0} = \frac{n_1}{n_1 + n_2}$

Wt. of glu cose = (w_1) = 18 g

Wt.of. water = W_2 = 90 g

Gmw of water = M_2 = 18 g

Num.of moles of glu cose $n_1 = \frac{W_1}{M_1}$

$= \frac{18}{180} = 0.1$

Num. of moles of water $n_2 = \frac{W_2}{M_2}$

Olympiad Class Work Book

IX-Chemistry (Vol-I)

$$= \frac{90}{18} = 5$$

$$\text{R.L.V.P} = \frac{n_1}{n_1 + n_2} = \frac{0.1}{0.1 + 5}$$

$$= \frac{1}{51}$$

$$\frac{\Delta_p}{P^0} = \frac{n_1}{n_1 + n_2}$$

W. of glucose $= (w_1) = 18$ g

W.of.water $= W_2 = 90$ g

Gmwof water $= M_2 = 18$ g

Number of moles of glucose $n_1 = \frac{W_1}{M_1}$

$$= \frac{18}{180} = 0.1$$

Number of moles of water $n_2 = \frac{W_2}{M_2}$

$$= \frac{90}{18} = 5$$

$$\text{R.L.V.P} = \frac{n_1}{n_1 + n_2} = \frac{0.1}{0.1 + 5}$$

$$= \frac{1}{51}$$

3. Option -1

Acc tohenry'slaw, $\frac{S}{P} = K_H$

$$S = 3.1 \times 10^{-2} \times 4$$

$$S = 0.124 \text{ mol/Lit}$$

4. Option -2

$$\text{V.P } \alpha \frac{1}{\text{B.P}}$$

5. Option -3

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Narayana Group of Schools

$\log P = -\frac{\Delta H}{2.303R} \times \frac{1}{T} + \log A = -\frac{\Delta H}{2.303R}$

$Y = m x + c$

7. Option-2

AcctoRaoul 'slaw

$S = K_{H.P}$

$S = 1.3 \times 10^{-3} \times 0.21$

$S = 2.7 \times 10^{-4}$ mol/Lit

$S = 8.6$mg/Lit

8. Option -1

$S = K_{H.P}$

$S = 2.3 \times 10^{-2} \times 1$

$S = 0.023 \text{mole} CO_2$ Per Lit

$\therefore$ 1 LitSolution -0.023 mole/ Lit

0.9 LitSolution - - ?

$0.023 \times 0.9 = 0.0207$ moleof CO_2

9. Option -2

$$\text{RLVP} = \frac{n_1}{n_1 + n_2}$$

$$\text{RLVP} = \frac{26/52}{26/52 + 74/74}$$

$$= \frac{26/52}{26/52 + 1}$$

$$= \frac{26}{52} \times \frac{52}{78}$$

$\Rightarrow 0.33$

10. Option-1

$$\text{RLVP} = \frac{3/60}{3/60 + 36/18}$$

$$= \frac{0.05}{2.05}$$

$= 0.024$ atm

$1 \text{ atm} - 760$ mm of Hg

$0.024 - ?$

$\Rightarrow 18.6$ mm of Hg

$\simeq 19$

11. Option -4

$P_0 = 90$ mm

$P_s = 81$ mm

$$\text{RLVP} = \frac{P_0 - P_s}{P_0}$$

$$= \frac{90 - 81}{90} = \frac{9}{90}$$

RLVP = 0.1

12. Option-1

$$\frac{P_0 - P_S}{P_0} = \frac{W_B}{M_B} \times \frac{M_A}{W_A}$$

$$\frac{0.0345}{0.645} = \frac{2.64 \times 74}{M_B \times 30}$$

$M_B = 121.7 \simeq 122$

13. Option-2

$$550 = P^0 \times \frac{1}{4} + P_y^0 \frac{3}{4}$$

$$550 = \frac{P_x{}^0 + 3P_y{}^0}{4}$$

$$2200 = P_x^0 + 3P_y^0 - - - 1$$

$$2200 = P_x^0 \frac{1}{5} + P_y^0 \times \frac{4}{5}$$

$$560 = \frac{P_x^0 + P_y^0 \times 4}{5}$$

$$2800 = P_x^0 + 4P_y^0 - - - -2$$

2-1

$$2800 = P_x^0 + 4P_y^0$$

$$-2200 = P_x^0 + 3P_y^0$$

$$P_y^0 = 600$$

$$2800 = P_x^0 + 4 \times 600$$

$$P_x^0 = 2800 - 2400$$

$$P_x^0 = 400$$

14. Mass $= K_H \cdot P$

$$K_H = \frac{2.2 \times 10^{-4}}{1.2}$$

$= 1.83 \times 10^{-4}$ g/atm

Solubility $= K_H \times P$

$= 1.83 \times 10^{-4} \times 10$

$= 1.83 \times 10^{-3}$ g/100 gH_2O

15. Option -2

$$\chi_{N_2} = \frac{P_{N_2}}{K_H} = \frac{0.987}{76480}$$

$= 1.29 \times 10^{-5}$

If n represents num of moleof N_2

In 1LitreH_2O or 55.6 moleH_2O, then

$$\chi_{N_2} = \frac{n}{n + 55.6} = \frac{n}{55.6}$$

$= 1.29 \times 10^{-5} (\because n << 55.6)$

$n = 1.29 \times 10^{-5} \times 55.6$ mol

$n = 7.17 \times 10^{-4}$ mol

n = 0.717mil limol

16. Option -4

$$\frac{P_0 - P_s}{P_0} = \chi_2$$

$$\frac{760 - 750}{760} = \chi_2$$

$$\frac{10}{760} = \chi_2$$

$$\chi_2 = \frac{1}{76}$$

26. KNO_3 is more soluble.

27. It is an exception.

Worksheet – 4

Level – I

1. B	2. D	3. B	4. D	5. B	6. C	7. D	8. C
9. A	10.A						

Level – II

1. B	2. B	3. C	4.A	5. 1	6. 2	7. 2	8. 4
9. A	10. D	11. B	12. 2	13. 3	14. 1	15. 1	16. 2
17. A, D	18. A,B,C,D	19. A	20. 1	21. 2	22. 3		
23. a – p,s b – p,r; c – p; d – p,r			24. 2,4	25.1,2,4	26. 3	27. 3	
28. D	29. a – p, b – q,r,s c – p d – r,s			30.1			

Worksheet – 5

Level – I

1. A	2. B	3. C	4. A	5. B	6. B	7. C
8. B	9. A					

Level – II

1. B	2. C	3. C	4. D	5. D	6. C	7. A	8. C
9. A	10. B	11. B	12. D	13. B	14. B	15. A	
16. A,D	17. B	18. B	19.C	20. 100.0790°C			
21. a – p;b – q;c – r;d – s							
22. A,B,D	23. A,C,D	24. A	25. C	26. B	27. a – p,q;b – r;c – s;d – t		
28. A,B,C	29. A	30. D	31. C	32. B			

2. $T_b - T_b^0 = k_b \times m$

$0.66 = 3.28 \times \frac{3.3}{M}$

$M = \frac{3.28 \times 3.3 \times 8}{0.66}$

$M = 131.2$

3.

$\Delta T_b = k_b \times \frac{10}{100} \times \frac{1000}{100}$

$k_b = \Delta T_b$

6. $\Delta Tb = k_b \times \frac{w}{m} \times \frac{1000}{w}$

$0.1 = k_b \times \frac{1.8}{180} \times \frac{1000}{1000}$

$k_b = 0.1 \times 10\ k_b = 1$

7. $\Delta T_b = k_b \times \frac{w}{m} \times \frac{1000}{w}$

$\Delta T_b = 0.52 \times \frac{12.5}{60} \times \frac{1000}{170}$

$= 0.52 \times \frac{12.5}{6} \times \frac{10}{17}$

$= \frac{0.52 \times 12.5 \times 10}{102}$

$= 0.63$

8. $m = ?, kf = 5.1$

$\Delta T_f = mk_f$

$$m = \frac{\Delta T_f}{k_f}$$
$$= \frac{0.8}{5.1}$$
$$= 0.157$$

9. $wt = 5\ kg$
$= 5 \times 1000$
$5000\ g$
$$0.3 = 1.86 \times \frac{x}{62} \times \frac{1000}{5000}$$
$$x = \frac{0.3 \times 62 \times 5}{1.86} = 50$$

10.
$m = 0.5$
$\Delta k_f = 1.86$
$\Delta T_f = ?$
$\Delta T_F = m \times \Delta k_f$
$= 0.5 \times 1.86$
$= 0.93^\circ C\ FP_{solution} = 0 - 0.93 = -0.93^\circ C$

11. $k_f = ?$
$\Delta T_f = 1.66^\circ C$
$$k_f = \frac{RT^2}{1000 L_f}$$
$$k_f = \frac{8.314 \times (289.6)^2}{1000 \times 180.75}$$
$$k_f \simeq 3.86$$

12. $\Delta T_f + \Delta T_b = (K_f + K_b)\left(\frac{W_a}{m_a} \times \frac{1000}{W}\right)$
$(0 - T_f) + (T_b - 1000) = m(K_f + K_b)$
$(-T_f + T_b) - 100 = m(K_f + K_b)$
$104 - 100 = m(2.38)$
$m = 1.68$
$$n \times \frac{1000}{500} = 1.68$$
$\Rightarrow n = 0.84$

13.
$\Delta T_b = k_b \times m$
$$0.1 = \frac{0.11}{M} \times \frac{1000}{15} \times 2.16$$
$$= \frac{0.11 \times 1000 \times 2.16}{1.5}$$
$$= \frac{237.6}{1.5}$$
$= 158.4$

14.

$$\Delta T_b = k_b \times m$$
$$0.13 = k_b \times \frac{10.8}{180} \times \frac{1000}{240}$$
$$k_b = \frac{0.13 \times 18 \times 24}{10.8 \times 10}$$
$$k_b = \frac{56.16}{108}$$
$$= 0.52$$

option 2

15. $100.18 - 100 = K_b \times 0.1$
$0.18 = K_b \times 0.1$
$K_b = \frac{0.18}{0.18}$
$= 1.8$
option = 1

20. $T_s - T^0_{solvent} = m \times k_b$
$T_s - 99.725°C = 0.513 \times 0.69$
$T_s = 100.0790°C$

24. $\Delta T_f = \frac{W}{M} \times \frac{1000}{W} \times K_f$
$o^0c - T_s = \frac{342}{342} \times \frac{1000}{1000} \times 1.86$
$T_s = -1.86°C$

25. $\Delta T_f = K_f \times m$
$K_f = \frac{1.86}{0.1} = 18.6$
$0.1 \times v + 0.3 \times v = M_3 \times 2v$
$M_3 = 0.2$
$\Delta T_f = K_f \times m$
$= 18.6 \times 0.2$
$= 3.72°C$

26. $\Delta T_f = \frac{W}{M} \times \frac{1000}{W_{gr}} \times K_f$
$K_f = \frac{\Delta T_f \times M \times W_{gr}}{W \times 1000}$
$K_f = 5.104$ K molarity $^{-1}$

30. $n_{napthalene} = 0.1$ mole
$n_{benzene} = 0.9$ mole
$\Delta T_b = 353$ K: $\Delta T_f = 278.5$ K
V.P = 670: $\Delta H = 10.67$KJ/M
i. amount of solifield benzene is
$\frac{P_0 - P_s}{P_s} = \frac{W_b}{m_b} \times \frac{m_a}{W_a}$
mass of solute $= 0.9 \times 78$
$= 70.2$
$P_0 = 760: P_s = 670$

$$\frac{760-670}{670}=\frac{0.1\times 78}{W_a}$$

$W_a = 50.06$ g

∴ Amount if solidifield benzeneis

$= 70.2 - 58.02$

$= 12.12$ g

31. $\Delta T_f = m \times K_f$

$$K_f = \frac{RT_f}{1000 \times L_v}$$

$L_v = 136.8$

$$K_f = \frac{8.314 \times 278.5 \times 278.5}{1000 \times 136.8}$$

$= 8.12$

Orginal solution $= 278.5 - 8.12$

$= 270.38$ K

32. $n_{\text{moles of Napthalene}} = 0.2$

$n_{\text{moles of Benzene}} = 0.8$

$\Delta T_f = ?$

$\Delta T_f = m \cdot k_f$

$$= 1000 \times 4.71 \times \frac{0.2}{0.8} \times \frac{1}{78}$$

$\Delta T_f = 15.16$

Worksheet – 6

Level – I

1. B	2. B	3. C	4. A	5. D	6. B	7. A
8. C	9. D	10.C	11. C			

Level – II

1. C	2. B	3. C	4. A	5. C	6. B	7. D	8. B
9. C	10. A	11. A	12. B	13. C	14. B	15. D	16. C
17. A	18. B	19. C	20. B	21. A	22. D	23. D	24. C
25. a - p; b - s; c - p,q d -r		26. C,D	27. B,C	28. A	29. D	30. B	
31. a – s;b – p ;c – r;d – q		32.A,B,C					

1. $(\Pi = MRT)$

$(\Pi \alpha M)$

2. $\Pi = CST$
   $C = \frac{m}{20} \Rightarrow \frac{60}{20} \Rightarrow 3$ (Mass of urea = 60.06.
   $c = 0.05$
   $s = 0.0821$
   $T = 27 + 273$
   $= 300k$
   $\Pi = CST$
   $= 0.05 \times 0.0821 \times 300$
   $5 \times 3 \times 0.0821$
   1.231 atm
3. T = constant
   p_1 = Glucose = 180 g
   p_2 = urea = 60 g
   p_3 = sucrose = 342 g
   $p_1 < p_2 < p_3$
   (M.M $\alpha \frac{1}{O.P}$)
4. $M(I.C = 0.1M$
   $T = 300$ K
   $\Pi = CST$
   $= 0.1 \times 0.0821 \times 300$
   $= 2.463bar$
5. $\% = 5\%$
   $\frac{w_1}{m_1} = \frac{w_2}{m_2}$
   $\frac{5}{342} = \frac{1}{x}$
   $0.0146 = \frac{1}{x}$
   $x = 68.4gr$
   $x = 68.49$ (or.
   $x = \frac{1}{0.146}$
   x = 68.5 g
6. $w = 5gr$
   m = 342
   $\Pi V = nRT$
   $\Pi = \frac{5}{342} \times 0.082 \times \frac{423}{0.1}$
   $\Pi = 5.07$
8.
   W = 30gr
   v = 1lr
   T = 293k
   Π = 3.2 atm

$s = ?$

$m.\ m = 228$

$\Pi = \frac{n}{v} ST$

$S = \frac{V\Pi}{nT}$

$\frac{1 \times 3.2}{293 \times \frac{30}{228}}$

$\frac{1 \times 3.2}{293 \times 0.13}$

$= 0.083009$

$\simeq 0.0821$

9. $M = 6\ g\ n_1 =$ glucose mole Where

$V = 1ltr$

$n_2 =$ Urea mole

$\pi = \frac{n}{V} RT\ \pi = \left(\frac{n_1}{V_1} + \frac{n_2}{V_2}\right) V_1 = V_2$

$\frac{18}{180} + \frac{6}{60}) \times 0.0821 \times 300 \quad \pi = (n_1 + n_2)\left(\frac{RT}{V}\right)$

$= 4.92 \quad \pi = \left(\frac{18}{180} + \frac{6}{60}\right)\frac{RT}{V}$

10. $V = 200\ cm^3 \Rightarrow \frac{200}{1000} = 0.2r$

$m = 1.26gr,$

$M.Wt =?$

$\pi = \frac{n}{v} RT$

$M.Wt = \frac{wt \times R \times T}{V \times \pi}$

$\frac{1.26 \times 0.0821 \times 300 \times 10^3}{0.2 \times 2.57}$

60.377×10^3

$60.377gr$

11. $V = 1Ir\ W = 18\ g$

$T = 293\ s = 8.3\ J\ K^{-1}$

$M.Wt = \frac{wt \times R \times T}{V \times \pi}$

$= \frac{18 \times 8.3 \times 293}{10^{-9} \times 2.414 \times 10^5}$

$= \frac{437742.2}{2.414} \times 10^5$

$= 18133.47 \times 10^5$

$= 181.33\ g$

12. $\pi = \frac{n}{V} RT$

$$1.26 \times 10^{-3} = \left(\frac{W}{M}\right)\frac{RT}{V}$$
$$1.26 \times 10^{-3} = \left(\frac{1.26}{M}\right)\frac{0.0831 \times 300}{0.3\text{lit}}$$
$$M = \frac{1.26 \times 0.0821 \times 300}{1.26 \times 10^{-3} \times 0.3}$$
$$M = \frac{0.0821 \times 100}{10^{-4}} = 83100$$

13. $\pi = 8.21$ atm
T = 310 K
g. mass = 180
volume = 1ltr
$\pi = \frac{n}{V}RT$
$\pi = \left(\frac{W}{M}\right)\left(\frac{RT}{V}\right)$
$W = \left(\frac{\pi MV}{RT}\right) \Rightarrow W = \frac{8.21 \times 180}{0.0821 \times 310}$
$W = \frac{1477.8}{25.451}$ $W = 58.00$ g

Worksheet – 7

Level – I

1. D
2. A
3. A
4. A
5. B
6. C
7. D
8. B
9. C
10. B

5. $\alpha = \frac{i-1}{n-1}[2]$
6. Option:2

$AlCl_3 \rightarrow Al^{+3} + 3\bar{Cl}[T_1]$
n = 4

$CaCl_2 \rightarrow Ca^{+2} + 2Cl^{-}[T_2]$
n = 3
$\therefore S_0T_1 > T_2$

9. Option:3
$NaSO_4 \rightarrow 2Na^{+} + So_4^{-2}$
$i = 1 + (n-1)\alpha$
$i = 1 + (3-1)\alpha$
$i = 1 + 2\alpha$

Level – II

1. A	2. C	3.C	4. B	5. 1	8. C	9. B	10. C
11. B	12. C	13. C	14. B	15. 3	16. B	17. C	18. A
19. A	20. B,C	21. B,C,D	22. C	23. 3	24.A		
25. a – q;b – q;c – p;d – s		26. A,B,D	27. A,B,C	28. 3	29. B	30. B	

31.a– p,q,s ; b – q; c – r; d – q 32. A,D

3. $K_3[Fe(CN)_6] \rightarrow$
$3\ K^+[Fe(CN)_6]^{-3}$
$m = 4$

5. Option-1
$\alpha = 0\%$
$\alpha = 0.8$
$i = 1 + (n - 1)\alpha$
$i = 1 + (1)0.8$
$i = 1 + 0.8$
$i = 1.8$
$\pi = CRT$
$\pi = iCRT$
$\pi = (1.8)(0.1)(0.0821)(300)$
$\pi = 4.43$
$H_3PO_4 \rightarrow 3H^+ + PO_4^{-3}$

8. $n = 4$
$i = 4$

11. Option : 2
$K_4[Fe(CN)_6]$
$4K^+ + [Fe(CN)_6]^{-4}$
$n = 5$
$i = 1 + (5 - 1.1$
$= 1 + 4$
$i = 5$

12. Option-3
$\alpha = 0.9$
$C = 0.002$
$\pi = CRT$
$i = 1 + (n - 1)\alpha$
$= 1 + (1)0.9$
$i = iCRT$
$\pi = (1.9)(0.002)(0.0821)300$
$\pi = 0.094$ bar

13. Option-3
$\Delta T_f = i \times k \times m$
$0.68 = i \times 1.86 \times 0.2$
$i = 1.83$
$\alpha = \frac{i - 1}{n - 1}$
$\alpha = \frac{1.83 - 1}{2 - 1}$
$\alpha = 0.83$
p83%

14. Option-2

15. Option-3

$$\frac{\pi_1^3}{\pi_2} = \frac{\mathrm{CRT_1}}{\mathrm{CRT_2}}$$
$$\frac{\pi_1}{\pi_2} = \frac{\mathrm{T_1}}{\mathrm{T_2}}$$
$$\frac{4}{\pi_2} = \frac{273}{546}$$
$$\pi_2 = 8$$

16. Hmt:

$$\pi \mathrm{v} = \frac{\mathrm{W}}{\mathrm{M}} \times \mathrm{ST} \times \mathrm{i}$$

18. option-1

$$\Delta Tf = i \times kf \times m$$
$$2 \times i \times 1.72 = \frac{20/172}{50/100}$$
$$i = 0.5$$

19. 1

$$\Delta \mathrm{T}f = \mathrm{K}f \times m$$
$$\Delta \mathrm{T}f = i \times \mathrm{K}f \times m$$
$$0 - \mathrm{T}f = (1 + 0.2) \times 1.86 \times 0.2$$
$$\Delta \mathrm{T}f = -0.45^0\mathrm{C}$$

2. Thermodynamics

Worksheet – 1

Level – I

1. B	4. A	7. A	10. C
2. A	5. B	8. A	
3. D	6. C	9. C	

Level – II

1) C	2) B	3) C	4) D	5) C	6) C	7) C
8) C	9) B	10) C	11) B	12) A	13) A	14) D
15) C	16) B	17) C	18) D	19) D	20) B,B,D	21) 5
22) C	23) A	24) C	25)a-s; b-p,r; c-q,r; d-r		26) A,B	27) 7

a. $A \rightarrow B$ (Temperature and pressure are constant)
$\therefore$ It is isothermal and isobaric process

2. $C \rightarrow D$ (Temperature and volume are constant)
$\therefore$ This process is isothermal and isochoric process

3. $D \rightarrow E$ (Temperatue and pressure are constant)

4. $E \rightarrow F$ (It is an adiabatic compression in which temperature increases from T_2 to T_1)

a. At stage ' A '
$V = 24\ T = 300\ K; n = 1 R = 0.0821 atm K^{-1}\ mol^{-1}$
$PV = nRT$
$$P = \frac{1 \times 0.0821 \times 300}{24} = 1.026\ atm$$

2. At stage ' B ' volume remains the same but temprature changes from 300k to 600k. Thus according to pressure law, the pressure will be doubled at B with respect to ' A '
Pressure of $B = 2 \times 1.026 = 2.052$ atm

3. At stage ' C ' temperature is 300k and volume is half that of stage A. Thes, according to bolyle's law, the pressure at c will be doubled with respect to A.
Pressure at $C = 2 \times 1.026 = 2.052$ atm.

Worksheet – 2

Level – I

1. B	4. B	7. A	10. B
2. B	5. B	8. A	11. D
3. A	6. D	9. C	

Level – II

1) C	2) B	3) B	4) B	5) A	6) C	7) B	8) B
9) B	10) A	11) B	12) A	13) D	14) D	15) B	16) A
17)A	18) A	19) A	20) C	21) B	22) D	23) B	
24) 1,2,34	25) C	26) C	27) C	28) B	29) D	30) C	31) 8

19. Given $2\ A_{(s)} + 2\ B_{(g)} \longrightarrow 2C_{(g)} + D_{(g)}$

Correction in the equation $2\ A_{(g)} + 2\ B_{(g)} \longrightarrow 2C_{(g)} + D_{(g)}$

$\Delta n = 3 - 4 = -1$

$\Delta H = \Delta U + \Delta nRT$

$-28000 = \Delta U - 1 \times 2 \times 300$

$\Delta U = -28000 + 600$

$= -27400\text{Cal}$

$= -27.4\text{Kcal}$

20. $CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(\ell)}$

$\Delta n = 1 - 3 = -2$

$\Delta H - \Delta U = -2 \times 2 \times 300 = -1200\text{cal}$

23. $C_2H_{4(g)} + 3O_{2(g)} \longrightarrow 2CO_{2(g)} + 2H_2O_{(\ell)}$

$\Delta n = 2 - 4 = -2$

$\Delta H = -1415000 - 2 \times 8.314 \times 300$

$= -1415000 - 4988.4$

$= -1419.988\text{KJ}$

Worksheet – 3

Level – I

1)A	4) C	7) A	9) D
2) B	5) D	8) D	10) D
3) C	6) D		

7. $C_P = ?\ C_V = 0.075\text{Cal/mol}$

Heat absorbed one mole of the gas is $= 40 \times 0.075 = 3\text{cal}$

$C_P = R$

$C_P - C_V = R(\text{ or })C_P = 5$

$C = \frac{5}{3}R = 1.66$ (Monoatomic)

Level – II

1) D	2) C	3) C	4) D	5) B	6) C	7) B	8) D
9) C	10) A	11) D	12) B	13) D	14) C	15) B,C	

16) A,D 17) C 18) A,B,D 19) C 20) A 21) D 22) C 23) D
24) A 26) A 26) B 26) A 27) 7 28) B

4. $\Delta H = \Delta E + \Delta nRT$
$\Delta H - \Delta E = \Delta nRT_1 : T_1 = 300\ K$
$\Delta H - \Delta E = \Delta nRTT_2 : T_2 = 350\ K$
$\therefore\ = 4 \times 8.3 \times 300 = 9960$
$= 4 \times 8.3 \times 350 = 11620$
$\Delta H - \Delta E$ at $350 = 11620$ J
$\Delta H - \Delta E$ at $300 = 9960$ J
$\Delta H - \Delta E = 1660$ J

6. Specific heat × atomic weight = 6.4
According to the dulong and Petit rule the "specificheat meultiplied by atomic mass" is always 6.4
atomic weight = 40
atomic $i + 4 = \frac{6.4}{12.52} = 0.5111$
atomicity won't be in fraction, so
Monoatomic

7. $q = m \cdot s\Delta T$
Given $q = 2137.5$
$T = (773 - 298) = 475k$
Specific heat = 0.13
m = mass of gold
$2137.5 = m \times 0.13 \times 475$
$m = \frac{2137.5}{0.13 \times 475} = \frac{2137.5}{61.75} = 34.615$
m = 34.61 g

12. $q = nC_V\Delta T = \frac{W}{M}C_V\Delta T = \frac{100}{18} \times 75 \times 2.4 = 1000J$

13. $C_P = C_V + R$ for diatomic gas, $C_V = \frac{5}{2}R, C_P = \frac{7}{2}R$

14. Heat required to raise the temperature from 40°C to 100°C at constant volume is given as $q = nC_V\Delta T$ where n is the number of moles. C_V is molar heat capacity and ΔT is the temparature change
$n_{Ar} = \frac{\text{Mass}_{Ar}}{\text{Molar Mass}_{Ar}} = \frac{40\ \text{g}}{40\ \text{g mol}^{-1}} = 1\ \text{mol}$
For monoatomic gases like argon, $C_V = 3\text{calmol}^{-1}$. Thus $q = (1)(3)(100 - 40) = 180\text{cal}$

23. $C_{(s)} + \frac{1}{2}O_{2(g)} \longrightarrow CO_{(g)}$
$\Delta n = 1 - \frac{1}{2} = \frac{1}{2}$ 0.5 R = 8.314
$\Delta H = \Delta U + \Delta nRT\ T = 298\ K$
$= 0.5 \times 8.31 \times 298$
$\Delta H - \Delta U = 1238.78$ J/mol

Worksheet – 4

Level – I

1. C
2. D
3. A
4. C
5. A
6. C
7. D
8. C
9. C
10. A

Level – II

1. A
2. A
3. A
4. A
5. D
6. C
7. C
8. A
9. D
10. A
11. C
12. D
13. B
14. C
15. B
16. B
17. A,C
18. B,C
19. A,D
20. A, B, C
21. A,C
22. B,C
23. A
24. A
25. A,C
26. 8
27. a-r; b-p; c-s; d-q
28. C
29. 7
30. $a-p,r,s; b-q,r,s; c-q,r; d-r,s$

1. Work done in Reversible Iso thermol expansion

$$W = -2.303nRT\log\left(\frac{V_2}{V_1}\right)$$

$$n = \frac{wt}{mwt} = \frac{16}{32} = \frac{1}{2} = 0.5 \text{ moles}$$

n = Number of moles of oxygen = 0.5

R = 8.314 J

T = 300 K

$V_1 = 5dm^3$

$V_2 = 25dm^3$

$= -2.303 \times 0.5 \times 8.314 \times \log\left(\frac{25}{5}\right)$

$\Rightarrow \log\left(\frac{25}{5}\right) = \log 25 - \log 5$

$= \log 25 = 1.40 \log\left(\frac{a}{b}\right) = (\log a - \log b)$

$= \log 5 = 0.7$

$= -2.303 \times 0.5 \times 8.314 \times (1.40 - 0.7) - 2{,}872.07 \times 0.7$

$= -2010.4$

$W = -2.01 \times 10^{-3}$ J

2. $W = -\Delta PV$

$\Rightarrow P = 1$ atm

$V_2 = 15\ V_1 = 3$

$W = -1(15 - 3)$

$= -1 \times 12$ lit - atm

$= -12$ lit - atm

1 atm = 101.32 J

$= -12 \times 101.32$ J

$= -1215.9$

$= -1215 \times 10^{-3}$ J

3. $W = nRT\ W = 2 \times 2 \times 300$

$n = \frac{112}{56} = 2\ = 1200$cal

$R = 2\text{cal} = \frac{1200}{1000} = 1.2\text{kcal}$

$T = 300\text{k}$

8. $W = -P\Delta V$

$P = 1 \times 10^5\ N/m^2$

$V_1 = 1 \times 10^{-3}\ V_2 = 1 \times 10^{-2}$

$W = -P(V_2 - V_1)$

Work done $W = -1 \times 10^5(1 \times 10^{-2} - 1 \times 10^{-3})$

$= -1 \times 10^5 \left(\frac{1}{100} - \frac{1}{1000}\right)$

$= -1 \times 10^5 \left(\frac{10 - 1}{1000}\right)$

$= -1 \times 10^5 \times \frac{9}{1000}$

$= -9 \times 10^5 \times 10^{-3}$

$= -9 \times 10^2 = -9 \times 100$

$W = -900$ J

Worksheet – 5
Level – I

1. D	5. D	9. A	13. D
2. A	6. B	10. B	14. A
3. A	7. D	11. B	
4. A	8. C	12. D	

3. $C_{(\text{graphite})} \longrightarrow C_{(\text{diamond})}; \Delta H = ?$

$C_{(\text{graphite})} + O_{2(g)} \longrightarrow CO_{2(g)}; \Delta H = -94\text{kcal}$

$C_{(\text{diamond})} + O_{2(g)} \longrightarrow CO_{2(g)}; -68\text{kcal} -$

Equation (1) - Equation (2)

$C_{(\text{graphite})} + O_{2(g)} \longrightarrow CO_{2(g)}; \Delta H = -94\text{kcal}$

$C_{(\text{diamond})} + O_{2(g)} \longrightarrow CO_{2(g)}; \Delta H = -68\text{kcal}$

$C_{(\text{graphite})} \longrightarrow C_{(\text{diamond})}; \Delta H = -26kcal$

4. $C + O_2 \longrightarrow CO_2; \Delta H = -94\text{kcal}$

$CO + \frac{1}{2}O_2 \longrightarrow CO_2; \Delta H = -67.7\text{kcal}$

$C + O_2 \longrightarrow CO_2 + \frac{1}{2}O_2; \Delta H = -26.3\text{kcal}$

5. Heat of formation 1 mole of $HI = \frac{51.9}{2}$
6. The standard enthalpy of CO_2 is -393.5kj/mole

Level – II

1) C	2) C	3) B	4) A	5) B	6) C	7) D
8) D	9) C	10) A	11) A	12) B	13) D	14) C
15) B	16) D	17) C	18) A	19)B,C,D	20) A	21) C
22) A	23) B	24) a-s; b-p,r; c-p,r; d-p,q			25) B, D	
26) a-p,r; b-p,r; c-q,s; d-p,r			27) A, D		28) A	

1. $\frac{2m+2}{2} = m + 1$

   $N_2 + 2O_2 \rightarrow 2NO_2; \Delta H = 66\text{KJ}$
2. $\therefore$ For one mole $NO_2 = \frac{66}{2} = 33\text{KJ}$
3. Eque-(1) - eque- (2)
4. Eque - (1) - eque - (2)
5. $78\text{gm}C_6H_6$ combustion 718 K.cal

   $39\text{gm}C_6H_6$ combustion ?

   $\frac{718\times39}{78} = 359$ K.cal
7. $2C_6H_{2(\ell)} + 15O_{2(g)} \longrightarrow 12CO_{2(g)} + 6H_2O(\ell)$

   one mole of Benzene molecule = 78gms of Benzene

   two moles of Benzene molecule = 78×2 gms of Benzene

   $2 \times 78g = 15 \times 22.4$ litre of O_2

   For 39gms of benzene

   $39g = \frac{15 \times 22.4 \times 39}{2 \times 78} = 84$

   For 39gm of benzene are required 84 litres of O_2 for complete combustion.
8. The heat of combustion for $C = -349\text{KJmole}^{-1}$

   The heat of combustion for $H_2 = -241.8\text{KJmole}^{-1}$

   The heat of combustion for $CH_4 = -906.7\text{KJmole}^{-1}$

   The heat of formation of $CH_4 \Rightarrow \Delta H_f = \sum\Delta H_R - \sum\Delta H_P$

   $\Delta H_f = (-349 + 2 \times (-241.8) - (-906.7))$

   $\Delta H_f = (-349 - 483.6) + 906.7$

   $\Delta H_f = -832.6 + 906.7$

   ΔH_f of $CH_4 = 74.1\text{KJ}$.
9. $CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O(\ell)$

   $\Delta H = \Delta E + \Delta n_{(g)}RT$

   $\Delta H - \Delta E = \Delta n_{(g)}RT; R = 2Cal$

   $\Delta n-= 1 - 3 = -2; T = 300K$

$\Delta H - \Delta E = -2 \times 2 \times 300$

$\Delta H - \Delta E = -1200\text{Cal}$

12. The enthalpies of formation of $CO_{2(g)} = -393.5\text{KJmole}^{-1}$

The enthalpies of firmation of $H_2O_{(\ell)} = -286\text{KJ mole}^{-1}$

The enthalpies of formation of $C_2H_{4(\text{g})} = +52.3\text{KJ mole e}^{-1}$

The enthaply change for the combustion of $C_2H_{4(\text{g})}$ is

$C_2H_{4(g)} + O_2 \longrightarrow 2CO_2 + 2H_2O$

$\Delta H = 2.\Delta H_{CO_2} + 2\Delta H_{H_2O} - \Delta HC_2H_4$

$\Delta H = 2 \times -393.5 + 2 \times -286 - 52.3$

$\Delta H = -787 - 572 - 52.3$

$\Delta H = -1411.3K^2$ mole $^{-1}$.

13. Enthalpy of condensation is always positive.

14. $2C + H_2 \longrightarrow C_2H_2$

molar enthalpies of combustion of $C_2H_{2(g)} = -1300\text{KJ mole}^{-1}$

molar enthalpies of combustion of $C = -394KJ$ mole $^{-1}$

molar enthalpies of combustion of $H_2 = -286KJ$ mole e^{-1}

$\Delta H = 2\Delta H_C + \Delta H_{H_2} - \Delta H_{C_2H_2}$

$\Delta H = 2 \times -394 + (-286) + 1300$

$\Delta H = -788 - 286 + 1300$

$\Delta H = -1074 + 1300$

$\Delta H = +226KJ$ mole $^{-1}$.

15. ΔH_f^0 for $CO_{2(g)} = -393.5\text{KJ mole}^{-1}$

ΔH_f^0 for $CO = -110.5K'$ mole $^{-1}$

ΔH_f^0 for $H_2O(\ell) = -241.8\text{KJ mole}^{-1}$

$CO_{2(g)} + H_{2(g)} \longrightarrow CO_{(g)} + H_2O_{(\ell)}$ is

$\Delta H = \left(\Delta H_{CO} + \Delta H_{H_2O}\right) - \Delta HCO_2$

$\Delta H = (-110.5 - 241.8) + 393.5$

$\Delta H = -352.3 + 393.5$

$\Delta H = 41.2\text{KJmole}^{-1}$.

16. Resonance energy=observed heat of combustion-Calculated heat of combustion.

Resonance energy $= 50.4 - 232$

$= -181.6\text{KJmole}^{-1}$

17. Combustion of scrose is

$C_{12}H_{22}O_{11_{(s)}} + 12O_{2_{(g)}} \longrightarrow 12CO_{2_{(g)}} + 11H_2O_{(g)}$

$\Delta n = n_p - n_r$

$\Delta n = 23 - 12 = 11$

The relation between heat of reaction at constant pressure ΔH and the heat of reaction constant.

Volume ΔE is

$\Delta H = \Delta E + \Delta nRT$

$R = 1.98\text{Cal} = 1.98 \times 10^{-3}\text{kcal}$

$T = 25 + 273 = 298k$

$\Delta H = -1348.9 + 1.98 \times 10^{-3} \times 298 \times 11$

$\Delta H = -1348..41\text{kcal/mole}$

Worksheet – 6

Level – I

1. C
2. C
3. A
4. C
5. C
6. C
7. C
8. A
9. B
10. C
11. C

8. $H^+ + OH^- \rightarrow H_2O \cdot \Delta H = -57.3 kj/mole$

Level – II

1) B	2) C	3) B	4) B	5) C	6) B	7) D	8) B
9) A	10) B	11) D	12) A	13) D	14) A	15) A	16) A
17) C	18) B	19) A	20) A	21) B	22) B	23) A	24)B
25) A	26) B	27) A, D	28) A	29) D			

$H_2SO_4 \rightarrow 2H^+ + SO_4^{-2}$

$2 \times (-57.3KJ) = -114.6\ KJ$

2. $H_2SO_4 \rightarrow 2H^+ + SO_4^{-2}$

   $98gm - - - 27.2\ K.cal$

   $9.8gm - - - 2.72\ K.cal$

3. H_2SO_4 is a dibasic acid

   Hence $2x = y$ K.cal

   $\therefore x = \frac{y}{2}$

4. 'HCN' is a weak acid

   $\therefore 57.3 - 12.1 = 45.2KJ$

5. $\Delta H = H_2 - H_1 = q_2 - q_1$

6. For weak acid & weak base neatralisation heat of neatralisation is always less than 57.3KJ

7. $0.2 \times 57.3KJ = 11.46KJ$

8. $0.2 \times 57.3KJ = 11.46KJ$

9. $n = \frac{500 \times 1}{1000} = \frac{1}{2}$ moles

   The heat released for $\frac{1}{2}$ mole is 1.36 k.cal

   $\therefore$ Per 1 mole it is 2.72 k.cal

   $I.E. = 13.6 - 2.72 = 10.98 k.cal/mole$

10. Acidic nature α Heat of neatralization

11. for H_2SO_4 $N = 0.5 \times 2 = 1$ N

    for KOH $N = 1$ N

12. $N_2 + 3H_2 \rightarrow 2NH_3$

    $945 + 3 \times 436 \rightarrow 6 \times 391$

    ΔH = bond energy of reactants - bond energy of products

13. Let bond enrgy of A_2 be x then bond energy AB is also x and bond energy of $B_2\ \frac{x}{2}$

    Enthalpy of formation of AB is - 100kj/mole $A_2 + B_2 \rightarrow 2AB$

$$\frac{1}{2}\ A_2 + \frac{1}{2}\ B_2 \rightarrow AB; \Delta = -100KJ$$

$$-100 = \left(\frac{1x}{2} + \frac{x}{4}\right) - x$$

$$-100 = \frac{2x + x - 4x}{4}$$

$$\therefore x = 400KJ$$

$$A_2 + B_2 \rightarrow 2AB$$

$$\frac{1}{2}\ A_2 + \frac{1}{2}\ B_2 \rightarrow AB; \Delta = -100\ KJ$$

$$-100 = \left(\frac{1x}{2} + \frac{x}{4}\right) - x$$

$$-100 = \frac{2x + x - 4x}{4}$$

$$\therefore x = 400KJ$$

$$A_2 + B_2 \rightarrow 2AB$$

$$\frac{1}{2}\ A_2 + \frac{1}{2}\ B_2 \rightarrow AB; \Delta = -100KJ$$

$$-100 = \left(\frac{1x}{2} + \frac{x}{4}\right) - x$$

$$-100 = \frac{2x + x - 4x}{4}$$

$$\therefore x = 400KJ$$

14. Bond energy is the amount of heat required/ evolved to break/form 1 mole bond.
15. Average energy of $C - H$ bond $= \frac{360}{4} = 90$

    $C_2H_6 C - H$ bonds are present energy required to $6C - H = 90 \times 6 = 540$ energy of $C - C = 620 - 540 = 80$
23. For max rise in temp; max neutralization of $H^+$and OH^-required if we take equal volume, all H^+(5 m-mole) will react with all $OH^-(5\ m - mole\ )$
24. The enthalpy of dissociation of the $O - H$ bond depends on the molecular species from which H-atom is being separated.

    $H_2O(g) \longrightarrow H(g) + OH(g);\ \Delta H^0 = 501.87\ kJ\ mol^{-1}$

    However to break $O - H$ bond in hydroxyl a different quantity of heat (423.38 kJ mol^{-1}) is $BE(O - H) = \frac{501.87 kJ mol^{-1} + 423.38 kJ mol^{-1}}{2} = 462.625\ kJ\ mol^{-1}$
25. In case of diatomic molecules, bond enthalpy and bond dissociation enthalpy are identical because each refers to the reaction

    $H_2(\ g) \longrightarrow 2H(g);\ BE(H - H) = 433.93\ kJ\ mol^{-1}$
26. $BE(S - S) = \Delta H_{vap}\ (s) - \Delta H_f^0$

$$= \Delta H_{vap}\ (s) - \left\{\Delta H_f^0 (C_2H_5\ S - SC_2H_5)\right\} - \Delta H_f^0 (C_2H_5 - S - C_2H_5)$$
$$= 222.80 - \{201.92 - (-147.23)\}$$
$$= 277.49\ kJ\ mol^{-1}$$

Worksheet – 7

Level – I

1. B	4. A	7. D	10. A
2. A	5. B	8. A	
3. C	6. C	9. C	

Level – II

1) A	2) D	3) A	4) A	5) A	6) A	7) A
8) C	9) C	10) D	11) B	12) C	13) A	14) D
15) B	16) B	17) B	18) a-4; b-2; c-1; d-3			
19) a-p,r,; b-r; c-p,r; d-p,q,s.			20) B	21) A	22) C	23) D
24) A, C	25) A,B,C,D		26) C	27) B	28) A	

1. 4gms of $H_2 = 2$ moles for 2 moles $H - H$ bond energy - 208k cal
   for 1 mole $\frac{208}{2} = 104$
2. For 1 mole water $\Delta H_{\text{freee }} - 1435\text{cal}$
   for 1 gram water $\Delta H_{\text{freee }} = -1435/18$
3. $\Delta H_{(\text{graphite})} - \Delta H_{\text{diamond}}$
4. $\Delta H_{\mathrm{H_2O}(g)} - \Delta H_{\mathrm{H_2O}(l)}$
5. Eq(1) teq (2)
   $-35.25 - (-9.11) = -26.14$ kJ
9. $\Delta H = HP - HR = 335.2 - (-207.6)^{\circ}$
10. Equation - (1) - equation - (2)
11. Equation - (1) - equation - (2)
12. $\Delta H = \Delta H_{\mathrm{MgSO_4}} - \Delta H_{\mathrm{MgSO_4 7H_2O}}$
13. Due to extensive solvation of F^-ions on account of smaller size, observed value of heat of neutralisation of HF appears more.
14. Enthalpy of formation of $\mathrm{NH_3} = -46$ kJ/mole
    $\therefore \mathrm{N_2} + 3\mathrm{H_2} \rightarrow 2\mathrm{NH_3}\ \Delta \mathrm{H}_f = -2 \times 46\text{kJmol}$
    Bond breaking is endothermic and Bond formation is exothermic
    Assuming ' x ' is the bond energy of $\mathrm{N-H}$ bond (kJmol^{-1})
    $\therefore 712 + (3 \times 436) - 6x = -46 \times 2$
    $\therefore x = 352$ kJ/mol
17. By adding the two given equations, we have
    $H_{2(g)} + \frac{1}{2}\mathrm{O}_{2(g)} \rightarrow H^+_{(aq)} + \mathrm{OH}^-_{(aq)}; \Delta H = -228.88$ kJ
    Here ΔH^0_f of $H^+_{(aq)} = 0$
    $\therefore \Delta H^0_f$ of $\mathrm{OH}^- = -228.88$ kJ

3. CHEMICAL KINETICS

Worksheet – 1

Level – I

1)D 2) A 3) B 4) C 5) A 6) C 7) C 8) A
9)D 10) D

1. $A \rightarrow B$

$$-\frac{d[A]}{dt} = \frac{d[B]}{dt}$$

Level – II

1)D	2)B	3)D	4)C	5)D	6)D	7)B	8)A
9)C	10)B	11)B	12)A	13)A	14)B	15)D	16)C
17)C	18)A,C	19)A,B	20)A,D	21)D	22)A	23)D	

24) a-p,s,t; b-q ; c-r ; d-s 25) A 26)a-q; b-p ; c-r ; d-s 27)A
28)a-p,q; b-p,r ; c-p,s ; d-p,

2. $\frac{1}{2}\ A \rightarrow 2\ B$

$$-2\frac{d[A]}{dt} = \frac{1}{2}\frac{d[B]}{dt}$$

$$-\frac{d[A]}{dt} = \frac{1}{4}\frac{d[B]}{dt}$$

4. $2\ A \rightarrow B$

$$\text{Rate} = -\frac{1}{2}\frac{d[A]}{dt} = \frac{d[B]}{dt}$$

(or)

$$-\frac{1}{2}\frac{\Delta[A]}{\Delta t} = \frac{\Delta[B]}{\Delta t}$$

5. $aA + bB \rightarrow cC + dD$

$$\text{Rate} = \frac{1}{a}\frac{d[A]}{dt} = \frac{-1}{b}\frac{d[B]}{dt} = \frac{1}{C}\frac{d[C]}{dt} = \frac{1}{d}\frac{d[D]}{dt}$$

12. $N_2 + 3H_2 \rightleftarrows 2NH_3$

$$-\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$$

$$\frac{d[H_2]}{dt} = -0.3 \times 10^{-1} MS^{-1}$$

$$-\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$$

$$-\frac{1}{3}(-0.3 \times 10^{-4}) = \frac{1}{2}\frac{d[NH_3]}{dt}$$

$$\frac{0.3 \times 10^{-4}}{3} = \frac{1}{2}\frac{d[NH_3]}{dt}$$

$$\frac{2 \times 0.3 \times 10^{-4}}{3} = \frac{d[NH_3]}{dt}$$

$$\Rightarrow \frac{d[NH_3]}{dt} = 2 \times 10^{-5} = 0.2 \times 10^{-4} Ms^{-1}$$

13. $N_2O_4 \rightarrow 2NO_2$

$$-\frac{d[N_2O_4]}{dt}=\frac{1}{2}\frac{d[NO_2]}{dt}$$
$$-\frac{(0.32-0.5)}{30}=\frac{1}{2}\frac{d[NO_2]}{dt}$$
$$-\frac{(0.18)\times 2}{30}=\frac{d[NO_2]}{dt}$$
$$\frac{0.36}{30}=\frac{d[NO_2]}{dt}\ 0.012=\frac{d[NO_2]}{dt}$$

14. $A+3B\rightarrow 2C$

$$-\frac{d[A]}{dt}=-\frac{1}{3}\frac{d[B]}{dt}=\frac{1}{2}\frac{d[C]}{dt}$$
$$-\frac{d[A]}{dt}=3\times 10^{-3}\ \text{mol}\cdot\text{litt}^{-1}\cdot\text{min}^{-1}.$$
$$3\times 10^{-3}=-\frac{1}{3}\frac{d[B]}{dt}$$
$$\frac{d[B]}{dt}=9\times 10^{-3}$$

15. $3\ A\rightarrow XB$

$$-\frac{1}{3}\frac{d[A]}{dt}=\frac{1}{x}\frac{d[B]}{dt}$$
$$\frac{d[B]}{dt}=\frac{2}{3}\frac{d[A]}{dt}$$
$$-\frac{1}{3}\left(-\frac{d[A]}{dt}\right)=\frac{2}{3x}\left(-\frac{d[A]}{dt}\right)$$
$$\frac{1}{3}=\frac{2}{3x}$$
$$x=2$$

16. $\text{Rate}=\frac{\text{changein concentration}}{\text{time}}$

$$\text{Rate}=\frac{-d[A]}{dt}$$
$$=\frac{-[A_2-A_1]}{dt}$$
$$=\frac{-[0.05-0.2]}{5}$$
$$=\frac{-[-0.15]}{5\text{min}}$$
$$=\frac{0.15}{5\text{min}}$$
$$=\frac{0.15}{5\times 60\text{sec.}}$$
$$\frac{0.15}{300}\ \text{mol l. Iit. sec}^{-1}.$$
$$=0.0005\ \text{mol. lit}^{-1}\cdot\text{sec}^{-1}.$$

17. $xA+yB\rightarrow zC$

$$-\frac{d[A]}{dt}=-\frac{d[B]}{dt}=1.5\frac{d[C]}{dt}$$
$$-\frac{d[A]}{dt}=-\frac{d[B]}{dt}=\frac{3}{2}\frac{d[C]}{dt}$$
$$-\frac{1}{3}\frac{d[A]}{dt}=-\frac{1}{3}\frac{d[B]}{dt}=\frac{1}{2}\frac{d[C]}{dt}$$
$$\therefore 3\ A+3\ B\rightarrow 2C$$
$$\therefore x=3, y=3, z=2$$

25. $2\,A + B \rightarrow 2C + 3D$

$\text{Rate} = -\frac{1}{2}\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \frac{1}{2}\frac{d[C]}{dt} = \frac{1}{3}\frac{d[D]}{dt}$

$\text{Rate} = -\frac{1}{2}\frac{d[C]}{dt}$

$\text{Rate} = \frac{1}{2} \times 1 = 0.5$

$$-\frac{1}{2}\frac{d[A]}{dt} = 0.5 \Rightarrow \frac{-d[A]}{dt} = 1$$
$$-\frac{1}{2}\frac{d[B]}{dt} = 0.5 \Rightarrow \frac{-d[B]}{dt} = 0.5$$
$$-\frac{1}{3}\frac{d[D]}{dt} = 0.5 \Rightarrow \frac{-d[D]}{dt} = 1.5$$

27. $\text{Rate} = \frac{1}{2}\frac{d[KI]}{dt}$

$= \text{Rate} = \frac{-d[x]}{dt}$

$\text{rate} = \frac{-d[\text{ final conc. } - \text{ initial conc. }]}{\text{Changein time}}$

$\text{rate} = \frac{-d[x_2 - x_1]}{dt}$

$\text{rate} = \frac{-1}{2}\left[\frac{0.05-2}{10 \text{ min.}}\right]$

$= \frac{-1}{2}\left[\frac{-0.15}{10 \times 60\text{sec}}\right]$

$= \frac{0.15}{1200} = \frac{15}{12} \times 10^{-4}$

Rate $= 1.25 \times 10^{-4}$ mol/ lit.sec .

Worksheet – 2

Level – I

1) D	2) C	3) A	4) B	5) C	6) C	7) C	8) D
9)A	10) B						

Level – II

1)B	2) A	3) B	4) A	5)B	6)C	7)C	8)D
9)A	10) A	11) B	12) B	13)C	14)D	15)A	16)A
17)A,B	18)A,B,C	19) C	20)C,D	21)a-q; b-p,s ; c-t ; d-r		22)B	23)B
24) C	25) B	26) C	27) B	28) B			

4. $\Delta H = (E_a)_f - (E_a)_b$

$\Delta H = 5$Kcal

$E_f = 15 - 5$

$E_b = 10$ Kcal.mol.

5. $\Delta H = 50$

$\Delta H = E_f - E_b$
$50 = 70 - E_b$
$\mathrm{E_b} = 70 - 50$
$\mathrm{E_b} = 20\mathrm{KJ} \cdot \mathrm{mol}^{-1}.$

8. $A \rightarrow C$
A.E. of forward = 25KJ.
A.E. of backward = 32KJ.
$\Delta H = ?$
$\Delta H = (E_a)_f - (E_a)_b$
$= 25 - 32$
$\Delta\mathrm{H} = -7\mathrm{KJ}.$

9. $\log \frac{K_1^1}{K_1} = \frac{E_1}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$
$\log \frac{K_2{}^1}{K_2} = \frac{E_2}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$
Since, $E_1 > E_2$
$\log \frac{K_1^1}{K_1} > \log \frac{K_2}{K_2}$
$\frac{K_1^1}{K_1} > \frac{K_2^1}{K_2}$

10. $K_1 = 1.6 \times 10^{-3} T_1 = 300K$
$K_2 = 3.2 \times 10^{-3} T_2 = 310K$
$\mathrm{E_a} = ?$
$\log \frac{K_2}{K_1} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 \times T_2}\right]$
$\log \frac{3.2 \times 10^{-3}}{1.6 \times 10^{-3}} = \frac{E_a}{2.303R}\left[\frac{310 - 300}{300 \times 310}\right]$
$\log 2 = \frac{E_a}{2.303 \times 2}\left[\frac{310 - 300}{300 \times 310}\right]$
$\log 2 \times 2.303 \times 2 = E_a\left[\frac{10}{310 \times 300}\right]$
$E_a = \frac{0.3010 \times 2.303 \times 2 \times 310 \times 300}{10}$
$E_a = 0.3010 \times 2.303 \times 310 \times 30$
$= 12893\mathrm{cal}.$
$E_a = 12.89 \simeq 13$ K.cal.

11. $T \rightarrow \infty$
$K = 3 \times 10^{-4}$
$E_a = 104.4\mathrm{KJ}$
$A = 6 \times 10^{14}$
$r = ?$
$K = Ae^{-\frac{E_a}{RT}}$

$$K = Ae^{-\frac{E_a}{R\infty}}$$
$$K = Ae^{0}$$
$$K = A$$
$$\frac{r_1}{r_2} = K\left(\frac{n}{\frac{V}{4}}\right)\left(\frac{n}{\frac{V}{4}}\right)$$

25. E_a = ?

$$\log(1.75) = \frac{E_a}{2.303 \times 1.987}\left(\frac{308-298}{298 \times 300}\right)$$
$$2.303 \times 1.987 \times \log(1.75) = E_a \frac{10}{298 \times 300}$$
$$2.303 \times 1.987 \times \log(1.75) = E_a \frac{10}{298 \times 300}$$
$$= \frac{2.303 \times 1.987 \times 298 \times 308}{10} \times \log(1.75)$$
10207.82Cal/mol.
$$\frac{10207.82}{1000} = 10.207\text{Kcal/mole.}$$

Worksheet – 3

Level – I

1)A	2)B	3)C	4)B	5)D	6)B	7)C	8)A
9)C	10)D						

Level – II

1)D	2)D	3)C	4)C	5)D	6)D	7)B	8)C
9)C	10)C	11)C	12)C	13)A	14)A	15)D	16)B
17)A,B,C	18)B,D	19)A,B,D	20)B	21)A	22)B	23)D	
24)a-3; b-4; c-2; d-1	25)4		26)B	27)C	28)A		

6. $E_a = 0$

$$\therefore K = Ae^{-\frac{E_a}{RT}}$$
$$K = A$$

10. For every 10°C/10 K temperature rate become double.

$$\frac{300K}{K_1} = \frac{280K}{K_2}$$
$$K_1 = 4K_2$$
$$K_2 = 0.25K_1$$

(or)

$$\frac{K_{t^0c+10}}{K_{t^0c}} = 2^n$$

$$n = \frac{T_2 - T_1}{10} \Rightarrow n = \frac{300 - 280}{10}$$

$$n = 2$$

$$\frac{K_1}{K_2} = 2^2$$

$$K_1 = 4K_2$$

$$K_2 = 0.25K_1$$

11. $\Delta H = E_f - E_b$

$$E_b = E_f - \Delta H$$

$$= 30 - (-20) = 50\text{KJ}$$

12. $\log K = \frac{-E_a}{2.303RT} + \log A$

13. For every 10°C rise in temp. rate becomes double.

$$\frac{K_{t^0c+10}}{K_{t^0c}} = 2^n$$

$$n = \frac{T_2 - T_1}{10} \Rightarrow \frac{80 - 40}{10}$$

$$n = 4$$

$2^4 = 16$

16×5 units

$= 80$ units

16. $\frac{d}{dt}\ln K = \frac{d}{dt}(\ln A) - \frac{d}{dt}\left(\frac{E_a}{RT}\right)$

$$= 0 - \frac{E_a}{R}\frac{d}{dt}(T^{-1})$$

$$= 0 - \frac{E_a}{R}(-1)T^{-1-1}$$

$$= 0 + \frac{E_a}{R}(T^{-2})$$

$$\frac{d}{dt}\ln K = +\frac{E_a}{RT^2} \quad \left(\because \frac{d}{dx}(x^n) = n \cdot x^{n-1}\right)$$

25. $\log \frac{K_2}{K_1} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$

$$\log(2) = \frac{E_a}{2.303 \times 8.314} \times \left[\frac{308 - 298}{308 \times 298}\right]$$

$$E_a = \frac{0.693 \times R \times 308 \times 298}{10}$$

$$= \frac{0.693R \times 308 \times 298}{10}$$

26.

$30 \rightarrow 40°\text{C} \rightarrow 50°\text{C} \rightarrow 60°\text{C} \rightarrow 70°\text{C} \rightarrow 80°\text{C}$

$a \quad 2a \quad 4a \quad 8a \quad 16a \quad 32a$

so 32 times increases.

27. every 10°C/10 K temp. reaction becomes double.

$K_2 = 2K_1$

$K_2 = 2 \times 0.02 = 0.04$

28. $\Delta H = E_f - E_b$

$-20 = 60 - E_b$

$E_b = 80\text{KJ./ mole}$

Worksheet – 4

Level – I

1)D	2)A	3)C	4)A	5)A	6)A	7)D	8)C

Level – II

1)C	2)A	3)B	4) D	5) D	6) D	7) B	8)A
9)D	10)B	11)B	12)B	13)D	14)A	15)B	16)B
17)A	18)A,B,C	19)B	20)C	21)C	22)B	23)a-r; b-s ; c-q ; d-p	
24)A,B,C	25)A,B	26)A	27)D	28)B			

2. Excess quantity will not consider in rate express on law.
   $r = k[A]^1[B]^0$
3. $r = k[A]^2[B]$
   Volume decreases concentration increases
   $r^1 = k[2a]^2[2B]$
   $r^1 = 8r$
4. $2P + Q \rightarrow$ products
   $r = K[P]^2[Q]$
   $r^1 = K[2P]^2[Q]$
   $r^1 = 4r$
5. (c) $= \frac{n}{v}$

$$\mathbf{V}_{\downarrow_{es}}\ C^{\uparrow^{es}}$$

$$\frac{V}{4} \Rightarrow 4C$$

$$r_1 = K\left(\frac{n}{V}\right)\left(\frac{n}{V}\right)$$

$$r_2 = K\left(\frac{n}{\frac{V}{4}}\right)\left(\frac{n}{\frac{V}{4}}\right)$$

$$\frac{r_1}{r_2} = \frac{K}{16}\frac{\left(\frac{n}{V}\right)\left(\frac{n}{V}\right)}{K\left(\frac{n}{V}\right)\left(\frac{n}{V}\right)}$$

$$\frac{r_1}{r_2} = \frac{1}{16}$$

$$r_2 = 16r_1.$$

6. $r_1 = K[SO_2]^2[O_2]$

$$r_1 = K\left(\frac{n}{V}\right)^2_{SO_3}\left(\frac{n}{V}\right)_{O_2}$$

$$r_2 = K\left(\frac{n}{3V}\right)^2_{SO_3}\left(\frac{n}{3V}\right)_{O_2}$$

$$\frac{r_2}{r_1} = \frac{K\left(\frac{n}{V}\right)^2\left(\frac{n}{V}\right)}{K\left(\frac{n}{3V}\right)^2\left(\frac{n}{3V}\right)}$$

$$\frac{r_2}{r_1} = 27\frac{\left(\frac{n}{V}\right)^2\left(\frac{n}{V}\right)}{\left(\frac{n}{V}\right)^2\left(\frac{n}{V}\right)}$$

$\frac{r_2}{r_1} = 27 \Rightarrow r_2 = 27r_1.$

7. $-\frac{1}{2}\frac{d[NH_3]}{dt} = \frac{d[N_2]}{dt} = \frac{1}{3}\frac{d[H_2]}{dt}$

$$-\frac{1}{2}\frac{d[NH_3]}{dt} = \frac{d[N_2]}{dt} = \frac{1}{3}\frac{d[H_2]}{dt}$$

$\frac{1}{2}\ K_1[NH_3]^2 = K_2[NH_3]^2 = \frac{1}{3}\ K_3[NH_3]^2$

$\frac{1}{2}K_1 = K_2 = \frac{K_3}{3}$

$3\ K_1 = 6\ K_2 = 2\ K_3$

10. Rate of reaction decreases with time due to decrease in concentration of reactants.

$r_1 = K(0.2)(0.2)$

$r_2 = K(0.1)(0.1)$

$\frac{r_1}{r_2} = \frac{K(0.2)(0.2)}{K(0.1)(0.1)}$

$\frac{r_1}{r_2} = 4 \Rightarrow r_2 = \frac{r_1}{4}$

12. Case(i): $r_1 = K[A]^n$

If concentration is doubled then the

Rate $= 4r_1 = K[2A]^n$

From Equation 1 and 2

$\frac{r_1}{4r_1} = \frac{K[A]^n}{K[2A]^n} \Rightarrow \frac{1}{4} = \left(\frac{1}{2}\right)^n$

$= 2^2 = 2^n$

n = 2.

Case(ii): $r_1 = K[A]^m \longrightarrow 1$

If concentration is tripled then the

Rate $= 9r_1 = K[3\ A]^m$

From Equation 1 and 2

$\frac{r_1}{9r_1} = \frac{K[A]^m}{K[3A]^m} \Rightarrow \frac{1}{9} = \left(\frac{1}{3}\right)^m$

$= 3^2 = 3^m \Rightarrow m = 2.$

13. $r_1 = K(C_A)^2(C_B)^{\frac{1}{2}}$

If concentration of A is doubled and concentration B is 4 times then the rate of reaction is 8 times.

$r_2 = K(2C_A)^2(4C_B)^{\frac{1}{2}}$

$r_2 = K4(C_A)^2 2(C_B)^{\frac{1}{2}}$

$r_2 = 8K(C_A)^2(C_B)^{\frac{1}{2}}; r_2 = 8r_1$

So, $C_A \times 2, C_B \times 4.$

15. $-\frac{1}{2}\frac{dA}{dt} = -\frac{dB}{dt}$

$$\frac{dB}{dt} = +\frac{1}{2}\frac{dA}{dt} = \frac{1}{2}\ K[\ A]^2[\ B]$$

17. $V \rightarrow \frac{V}{3} \Rightarrow [C] = 3[C]$
$r_1 = K[A][B]^2$
$r_2 = K[3A][3B]^2$
$r_2 = 27r_1$

19. $r = K[A]^{-1B}]^2$
$A + 2\ B \rightarrow P$
$-\frac{dA}{dt} = -\frac{1}{2}\frac{dB}{dt} \Rightarrow -2\frac{dA}{dt} = -\frac{dB}{dt}$

20. $-\frac{d[A]}{dt} = -\frac{1}{3}\frac{d[B]}{dt}$
$\frac{d[B]}{dt} = 3\frac{d[A]}{dt} = 3\ K[\ A][B]^3$

21. $\frac{dx}{dt} = K[A]^a[B]^b$
rate $= K[A]^1[B]^2$
$A + 2\ B \rightarrow$ Pr oducts
$\frac{-dA}{dt} = -\frac{1}{2}\frac{dB}{dt}$(or) $-2\frac{dA}{dt} = -\frac{dB}{dt}$

25. $-\frac{dA}{dt} = -\frac{1}{3}\frac{dB}{dt} \Rightarrow \frac{dB}{dt} = 3 \times K_1[\ A][B]^3$

26. $r = K[A]^n[B]^m$
$r^1 = K[2A]^n\left[\frac{B}{2}\right]^m$
$r^1 = r2^{n-m}$

Worksheet – 5

Level – I

1) C 2) D 3) A 4) A 5) D 6) C 7) C 8) D
9) D

1. $\left(\frac{\text{mole}}{\text{lt}}\right)^{1-n} \text{sec}^{-1}$
mole $\times$ lt^{-1} time $^{-1}$

Level – II

1)D	2)B	3)A	4)A	5)C	6)A	7)B	8)A
9)A	10)A	11)B	12)B	13)A	14)A	15)A	16)D
17)B	18)A	19)A,B	20)B,C	21)B	22)A	23)B	24)A
25)a-q; b-r ; c-s ; d-p,q	26)A	27)A,B,C	28)a-q; b-p ; c-p ; d-r,s				

1. $r_1 = K[A]^n$
$4r_1 = K[2A]^n$
From 1 and 2
$\frac{4r_1}{r_1} = \frac{K[2A]^n}{K[A]^n}$
$\frac{4}{1} = (2)^n\left(\frac{A}{A}\right)^n$
$2^2 = 2^n \Rightarrow n = 2$

5. $K = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right)[a = 100, x = 90]$

$t_{90\%} = \frac{2.303}{K}\log\left(\frac{100}{10}\right)$

$t_{90\%} = \frac{2.303}{K}\log 10$

$t_{90\%} = \frac{2.303}{K}\cdot 1$

$t_{90\%} = \frac{2.303}{K}$

6) $t_{\frac{1}{2}} = 15$ minutes

$$0.1 \xrightarrow{t_{\frac{1}{2}}} 0.5 \xrightarrow{t_{\frac{1}{2}}} 0.025$$

$t = t_{\frac{1}{2}} + t_{\frac{1}{2}} = 30$ min

7. order $= \frac{3}{2} - 1 = \frac{1}{2}$

8. $K = \frac{2.303}{5}\log\left(\frac{100}{80}\right) = 0.00446$

$$K = \frac{0.693}{t_{1/2}}; K = \frac{0.693}{0.00446}$$

$t = 155.38$

9. $r_1 = K[A]^n$

if $[A] = [3A]$

Then $9r_1 = K[3A]^n$

from 1 and 2

$\frac{9r_1}{r_1} = \frac{K[3A]^n}{K[A]^n}; \frac{9}{1} = \frac{[3]^n[A]^n}{[A]^n}$

$3^2 = 3^n$

n = 2.

11. $t_{\frac{1}{2}} = 6.93$

$t_{99\%} = 2t_{90\%} = 2 \times \frac{10}{3} t_{50\%}$

$= \frac{20}{3} \times 6.93 = 46.2$

12.

$$A \xrightarrow{t_{\frac{1}{2}}} \frac{A}{2} \xrightarrow{t_{\frac{1}{2}}} \frac{A}{4} \xrightarrow{t_{\frac{1}{2}}} \frac{A}{8} \xrightarrow{t_{\frac{1}{2}}} \frac{A}{16}$$

No. of half life $= \frac{\text{Total time}}{\text{Ha If life}}$

$n = \frac{60}{15} = 4$

Theamount of substance left $= \frac{\text{Conc.}}{2^n}$

$= \frac{[A]}{2^4} = \frac{[A]}{16}$

13. $t = \frac{2.303}{K}\log\frac{a}{a-x}$

$$t_{\frac{1}{4}} = \frac{2.303}{K}\log\frac{\frac{1}{3}}{4}$$
$$= \frac{2.303}{K}\log\frac{4}{3}$$
$$= \frac{0.2877}{K} = \frac{0.29}{K}$$

15. $t_{75\%} = 2 \times t_{50\%}$
$t_{75\%} = 1386$
$1386 = 2 \times t_{50\%} t_{50\%} = 693$
$t_{\frac{1}{2}} = \frac{0.693}{K} = K = \frac{0.693}{693} = 10^{-3}$.
16) $r = K[A]^x[B]^Y$

$$r^1 = K[2A]^x\left(\frac{B}{2}\right)^Y$$
$$r^1 = r[2^{x-y}]$$

17. $K = \frac{2.303}{10}\log\frac{50}{40}$
$K = \frac{2.303}{10}\log 1.25$

Worksheet – 6

Level – I

1)A	2)A	3)A	4)C	5)C	6)B	7)B	8)A
9)C	10)D						

8. $K = 6.2 \times 10^{-4}$

$t_{\frac{1}{2}} = ?$

$$t_{\frac{1}{2}} = \frac{0.693}{6.2 \times 10^{-4}} = 1117.7\text{sec}$$

Level – II

1)B	2)C	3)B	4)B	5)B	6)A	7)C	8)A
9)D	10)C	11)B	12)C	13)B	14)B	15)B	16)B
17)C	18)D	19)B,C	20)D	21)D	22)B		
23)a-q; b-p ; c-r ; d-r		24)A,B,C,D	25)A,B,D	26)a-q; b-r ; c-p,r ; d-r,s		27)C	28)C

1. $t_{\frac{1}{2}} = \frac{0.693}{K}$

$$K = \frac{0.693}{t_{\frac{1}{2}}}$$

$K = \frac{0.693}{10}$
$K = 0.0693$
$K = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right)$

$$t = \frac{2.303}{0.693}\log\left(\frac{0.08}{0.02}\right)$$

$$t = \frac{2.303 \times 2\log\ 2}{0.0693}$$

$$= \frac{2 \times 2.303 \times \log\ 2}{0.0693}$$

$$= 2 \times \frac{0.693}{0.0693}$$

$$= 2 \times 10$$

$$= 20 \text{ min.}$$

2. $t_{\frac{1}{2}} \alpha \frac{1}{a^{n-1}}$

3. $10^{-2} = K \times 0.2$

$$K = \frac{10^{-2}}{0.2}$$

$$K = \frac{1}{2} \times 10^{-1}$$

$$= 0.5 \times 10^{-1} = 0.05$$

$$t_{\frac{1}{2}} = \frac{0.693}{0.05}$$

$$= 73.6 = 14 \text{ s}$$

10. $t_{\frac{1}{2}} \propto \frac{1}{a^{n-1}}$ $t_{\frac{1}{2}} \propto a$

$$\frac{t_{\frac{1}{2}}^{1}}{t_{\frac{1}{2}}^{2}} = \left[\frac{a_2}{a_1}\right]^{n-1}$$

$$\frac{46}{92} = \left(\frac{0.2}{0.4}\right)^{n-1}$$

$$\Rightarrow \frac{1}{2} = \left(\frac{1}{2}\right)^{n-1}$$

$$2 = 2^{n-1}$$

$$n - 1 = 1$$

$$n = 2.$$

12. $T_{87.5\%} = 3t_{50\%} = 3\text{hrs}$

13. $t_{75\%} = 2 \times t_{50\%}$

$$32 = 2t_{50\%}$$

$$t_{50\%} = \frac{32}{2}$$

$$t_{50} = 16$$

14. $t_{99.99\%} = 10t_{50\%} = 10$ hour

15. $t_{93.75\%} 4t_{50\%}$

$$50 = 4t_{50\%}$$

$$t_{50\%} = \frac{50}{4} = 12.5 \text{ mm}$$

16. $t_{99\%} = 2t_{90\%}$

$$= 2 \times \frac{10}{3} t_{50\%}$$

$$= 2 \times \frac{10}{3} \times 10 = \frac{200}{3} = 66.6 \text{ years}$$

17. $t_{\frac{1}{2}} = 14.3$ days (24 hours/ 1 day)(3600 sec/ lhour) = 1,235,520sec

$K = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{1,235,520_s} = 5.61 \times 10^{-7}\ S^{-1}$

$[^{-32}P]\sqrt{[^{32}P]_6} = 0.05$

$t = \left(\frac{-1}{k}\right)$ in $0.05 = \left(\frac{-1}{5.61\times10^{-7}\ S^{-1}}\right)$

in $0.05 = 5.34 \times 10^7\ S$

∵ Converting into days 61.8 days.

18. $-\frac{1}{2}\frac{dA}{dt} = -\frac{dB}{dt} = \frac{dC}{dt}$

$2\frac{dC}{dt} = \frac{dA}{dt}$

19. $R = K[SO_2][SO_3]^{-\frac{1}{2}}$

20. $r = K[A]^2$

$2^2 = 4$

∴ The value n is 2 .

Worksheet – 7

Level – I

1)C	2)C	3)D	4)C	5)A	6)A	7)B	8)A
9)D	10)B						

Level – II

1)B	2)C	3)C	4)D	5)B	6)B	7)A	8)D
9)A	10)A	11)C	12)B	13)B	14)A	15)B	16)B
17)D	18)a-r; b-s ; c-t ; d-p		19)B	20)C	21)B		
22)a-r; b-p ; c-s ; d-t		23)A,B,C	24)A	25)A,B,D	26)A	27)A	28)B

3. $r = K[A][B]^2, r^1 = K[A][2B]^2 \Rightarrow r^1 = 4r$
4. I_2 not appearing in rate expression law
5. collision probability is low
6. $[A] = k_c{}^{1/2}[A_2]^{1/2}$

Thus,

$r = K \cdot K_c^{1/2}[A_2]^{1/2}[B_2] = K^1[A_2]^{1/2}[B_2]$

Therefore the order of reaction will be 1.5

10. $r = K[H_2O_2]$

order = 1

12. $O_3 \xrightarrow{K_c} O_2 + [O]$ (fast)

$O_3 + [O] \rightarrow 2O_2$ (slow)

$r = K[O_3][O]$

$K_c = \frac{[O_2][O]}{[O_3]} \Rightarrow [O] = \frac{K_c[O_3]}{[O_2]}$

$r = K \times K_C[O_3]^2[O_2]^{-1}$

$r = K^1[O_3]^2[O_2]^{-1}$

19. $R = K[A][B]^2$

Excess quantity will not count

$r = K[A]^0[B]^2$

Order = 2

20. $2A \xrightarrow{K_1} A_2 \Rightarrow K_1 = \frac{[A_2]}{[A]^2}$

$[A_2] = K_1[A]^2$

$A_2 + C \xrightarrow{K_2}$ Products

$$r = K_2[A]^2[C]$$

$$r = K_1K_2[A]^2[c]$$

25. $NO + Br_2 \xrightarrow{k_1} NOBr_2 \Rightarrow K_1 = \frac{[NOBr_2]}{[NO][Br_2]}$

$NOBr_2 + NO \xrightarrow{k_2} 2NOBr$

$\downarrow$

$[NOBR_2] = K_1[NO][Br_2]$

$r = K_2[NOBr_2][NO]$

$r = K_1K_2[NO]^2[Br_2]$

26. $0_3 \rightleftharpoons 0_2 + [0]$

$0_3 + [0] \xrightarrow{k_2} 2O_2$ slow

rate $= K_1K_2[O_3]^2[O_2]^{-1}$

Total order = +1

28. 0_2 showing negative order.

4. CHEMICAL EQUILIBRIUM

Worksheet – 1

Level – I

1. D 2. B 3. A 4. B 5. B 6. B

Level – II

1. D	2. D	3. A	4. D	5. D	6. B	7. C	8. C
9. C	10. A	11. D	12. B	13. C	14. D		
15. Only one product AgCl	16. A,D	17. A,D	18. A	19. C	20. A		21. B
22. a-r : b-p: c-q : d-s	23. C	24. A	25. A	26. a-s : b-q: c-r : d-s			

Worksheet – 2

Level – I

1. A 2. D 3. B 4. B 5. D 6. C 7. B
8. D 9. B

2. Active mass $= \frac{\text{w}}{\text{mwt}\times\text{V(lit)}}$

$= \frac{64}{128 \times 2} = 0.25$

3. $A_{(g)} + 2\ B_{(g)} \rightleftharpoons C_{(g)}$

$K_C = \frac{[C]}{[A][B]^2}$

$= \frac{0.216}{0.06 \times (0.12)^2}$

$= \frac{0.216}{0.06 \times 0.0144} = 250$

4. $A + B \rightleftharpoons C + D$

$K_C = \frac{[C][D]}{[A][B]}$

$= \frac{0.4 \times 1.0}{0.5 \times 0.8} = \frac{0.4}{0.4} = 1$

Level – II

1. D	2. D	3. D	4. C	5. C	6. D	7. C	8. B
9. B	10. D	11. A	12. C	13. A	14. A	15. C	16. A,B
17. 4	18. 2	19. 3	20. 1	21. 1,3,4	22. a- s; b-r ; c-p; d-q		
23. 2	24. 1	25. a- r,t; b-p,t ; c-q; d-s,t					

2.

$A + B \rightleftharpoons C + D$

Initial 1 1 0 0

at equilibrium

$$1\text{-}x \quad 1\text{-}x \quad 0.8 \quad 0.8$$

$$K_C = \frac{[C][D]}{[A][B]} = \frac{0.8 \times 0.8}{0.2 \times 0.2} = \frac{0.64}{0.04} = 16$$

8.

Initial

$H_2 + I_2 \rightleftharpoons 2HI$

at equilibrium

$0.3 \quad 0.3 \quad 0$

$0.3 - x \; 0.3 - x \; 2x$

$$K_C = \frac{[HI]^2}{[H_2][I_2]}$$

$$64 = \frac{\left[\frac{2x}{10}\right]^2}{\left[\frac{0.3-x}{10}\right]\left[\frac{0.3-x}{10}\right]}$$

$$64 = \frac{(2x)^2}{(0.3-x)(0.3-x)}$$

Squaring on both sides

$$8 = \frac{2x}{0.3-x} \Rightarrow 8(0.3-x) = 2x$$

$$2.4 - 8x = 2x$$

$$2.4 = 10x \Rightarrow x = 0.24$$

Then amount of

unreacted $I_2 = 0.3 - 0.24 = 0.06$

9.

$$R = K[NO]^2[O_2]$$

$$R = K\left[\frac{NO}{1/4}\right]^2\left[\frac{O_2}{1/4}\right]$$

$$R^1 = 64\ K[NO]^2[O_2]$$

$$R^1 = 64R$$

10. $R = K[A]^r[B_2]$

1.6×10^{-4} mole/lit/sec $= K(0.5)^r(0.5)$

$$K = \frac{1.6 \times 10^{-4}\ \text{mole/lit/sec}}{(0.5\ \text{mole/lit})^2(0.5\ \text{mole/lit})}$$

11.

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

initial

200

at equilibrium

2-x x x

1.2 0.8 0.8

Concentration $= \frac{n}{v} = \frac{1.2}{2} \; \frac{0.8}{2} \; \frac{0.8}{2}$

$$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{0.4 \times 0.4}{0.6} = 0.266$$

12. $N_2O_4 \rightleftharpoons 2NO_2$

$$K_C = \frac{[NO_2]^2}{[N_2O_4]} = \frac{\left(\frac{2 \times 10^{-3}}{2}\right)^2}{\frac{0.2}{2}}$$

$$= 1 \times 10^{-3} \times 10^{-3} \times 10$$

$$= 1 \times 10^{-5}$$

13. i) $R_1 = K_1[Cl_2O]^2$,

$$R_2 = K_2\left[\frac{1}{2}Cl_2O\right]^2 = K_2\frac{1}{9}[Cl_2O]^2$$

$$R_2 = \frac{1}{9}R_1$$

ii) $4^n = 2\ 2^{2n} = 2^1\ 2n = 1n = \frac{1}{2}$

iii) $R_1 = K_1[Cl_2O]^2$

$R_2 = K_1 3^2[Cl_2O]^2\ R_2 = 9R_1$

14. $H_2 + I_2 \rightleftharpoons 2HI$

$$K_C = \frac{[HI]^2}{[H_2][I_2]}$$

$$= \frac{(42.85)^2}{6.34 \times 4.02} = \frac{1836.1225}{25.4868} = 72.04$$

15. $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)}$

$$K_P = \frac{P_C}{P_A \times P_B}$$

$$K_P = \frac{0.30}{0.10 \times 0.15} = 20$$

$$20 = \frac{P_C}{(0.1)^2 \times (0.15)^2} = 1.2 \text{ atm}$$

23. $2H_2\ S_{(g)} \rightleftharpoons 2H_{2(g))} + S_{2(g)}$

$$K_C = \frac{[H_2]^2[S_2]}{[H_2\ S]^2} = \frac{(0.1)(0.4)}{(0.5)} = 0.008$$

Worksheet – 3
Level – I

1. B 2. C 3. C 4. C 5. B 6. A

4. $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)}K_1$

If we reverse the equation and multiply with 2 , then

$$2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}K_2 = \frac{1}{K_1^2}$$

5. $2C_{(s)} + O_{2(g)} \rightleftharpoons 2CO_{(g)}$

$$P_{O_2} = 2.0\ P_{CO} = 4.0\ P_C = 1$$

$$K_P = \frac{P_{CO}^2}{P_{O_2}} = \frac{4^2}{2} = 8$$

Level – II

1. A	2. B	3. A	4. B	5. C	6. D	7. A	8. A
9. D							
10. A	11. D	12. C	13. A	14. D	15. D	16. B	17. A
18. D							
19. a-s, b-p, c-q, d-r			20. B,D	21. C	22. a-s, b-r, c-p, d-q		
23. A	24. D	25. D	26. a-s, b-r, c-q, d-p				

3. $H_2 + I_2 \rightleftharpoons 2HI$

$$[H_2] = 0.1\ [I_2] = 0.1$$
$$[HI] = 0.8$$
$$K_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{0.8^2}{0.1 \times 0.1} = 64$$

4. $A_{(g)} + 2\ B_{(g)} \rightleftharpoons C_{(g)}$

$$[A] = 0.06\ [B] = 0.12$$
$$[C] = 0.216$$
$$K_C = \frac{[C]}{[A][B]^2} = \frac{0.216}{0.06 \times 0.12^2} = 250$$

6. $A + B \rightleftharpoons C + D$

If $[A] = x [B] = x$, then $[C] = 2x$ and $[D] = 2x$ At equilibrium

$$K_C = \frac{[C][D]}{[A][B]} = \frac{2x \times 2x}{x \times x} = \frac{4x^2}{x^2} = 4$$

8. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

$\Delta n = 2 - 4 = -2, R$

$= 0.0821$ litre $\cdot$ atm $\cdot$ mol^{-1} K^{-1}, T = 800 K

$K_P = K_C(RT)^{\Delta n}$

$$\frac{K_P}{K_C} = (RT)^{\Delta n} = (0.0821 \times 800)^{-2}$$

$$= \frac{1}{(0.0821 \times 800)^2} = 2.3 \times 10^{-4}$$

10. $2\ A_{(g)} + 2\ B_{(g)} \rightleftharpoons C_{(g)}$

$[A] = 0.06$

$[C] = 0.216$

$[B] = 0.12$

$$K_C = \frac{[C]}{[A]^2[B]^2} = \frac{0.216}{0.06^2 \times 0.12^2} = 4166$$

12. $K_b = 7.5 \times 10^{-4}$

$K_c = 1.5$

$K_f = ?$

$K_c = \frac{K_f}{K_b}$

$K_f = K_c \times K_b = 1.5 \times 7.5 \times 10^{-4}$

$= 1.125 \times 10^{-3}$

13. $3C_2H_2 \rightleftharpoons C_6H_6$ $K_C = 4$ at TK

$[C_2H_2] = 0.5$ $[C_6H_6] = ?$

$K_C = \frac{[C_6H_6]}{[C_2H_2]^3} \Rightarrow [C_6H_6] = K_C \times [C_2H_2]^3$

$= 4 \times 0.5^3 = 0.5$

14.

$CH_3COOC_2H_5 + H_2O \rightleftharpoons$

$CH_3COOH + C_2H_5OH$

$K_f = 1.1 \times 10^{-2},\ K_b = 1.5 \times 10^{-3}$

$K_c = \frac{K_f}{K_b} = \frac{1.1 \times 10^{-2}}{1.5 \times 10^{-3}} = 7.33$

15. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

$T = 500°C = 500 + 273 = 773$ K $K_P = 1.44 \times 10^{-5}$

$R = 0.0821$ lit.atm. $mol^{-1}k^{-1}$

$K_p = K_C(RT)^{\Delta n}$

$K_C = \frac{K_p}{(RT)^{\Delta n}} = \frac{1.44 \times 10^{-5}}{(0.0821 \times 773)^{-2}}$

20. $2H_2S_{(g)} \rightleftharpoons 2H_{2(g)} + S_{2(g)}$

$\Delta n = 3 - 2 = 1$

$[H_2S] = 0.5$ $[H_2] = 0.1$

$[S_2] = 0.4$

$K_C = \frac{[H_2]^2[S_2]}{[H_2S]^2} = \frac{0.1^2 \times 0.4}{(0.5)^2}$

$= 0.008$ or 8×10^{-3} mol/ litre

Worksheet – 4

Level – I

1. B 2. D 3. D 4. B 5. D 6. C 7. A 8. B
9. C 10. C

9.

$H_2O_{(g)} + CO_{(g)} \rightleftharpoons$

$H_{2(g)} + CO_{2(g)} \rightarrow k_1 = 0.44$

$2H_2 + 2CO_2 \rightleftharpoons 2CO + 2H_2O \rightarrow k_2$

$k_2 = \frac{1}{(k_1)^2}$ $k_1 = 0.44$

$= \frac{1}{(0.44)^2} = \frac{1}{0.1936} = 5.16$

10. $A_2 + B_2 \rightleftharpoons 2AB \rightarrow k_1$

$$A_2 + B_2 \rightleftharpoons 2AB \times \frac{1}{3}$$

$$\frac{1}{3}\ A_2 + \frac{1}{3}\ B_2 \rightleftharpoons \frac{2}{3}AB \rightarrow k_2$$

$$k_2 = (k_1)^{\frac{1}{3}}$$

Level – II

1. A	2. C	3. C	4. B	5. D	6. B	7. A	8. C
9. C	10. B	11. D	12. D	13. A	14. C	15. A,B	
16. B	17. A	18. B	19. 2	20. a-q; b-a; c-s,d-r,e-t			
21. B	22. A	23. A	24. B,D	25. A,B,C			

1. $A + B \rightleftharpoons C \rightarrow k_1$
   $B + C \rightleftharpoons P + Q \rightarrow k_2$
   $A + 2\ B \rightleftharpoons P + Q \rightarrow k_3$
   $k_3 = k_1 \times k_2$
2. $N_2 + O_2 \rightleftharpoons 2NO \rightarrow k_1$
   $2NO + O_2 \rightleftharpoons 2NO_2 \rightarrow k_2^2$
   $N_2 + 2O_2 \rightleftharpoons 2NO_2 \rightarrow k_3$
   $k_3 = k_1 \times k_2^2$
3. $A \rightleftharpoons C \rightarrow k_3$
   $A \rightleftharpoons B \rightarrow k_1$
   $B \rightleftharpoons C \rightarrow k_2$
   $A \rightleftharpoons C$
   $k_3 = k_1 \times k_2$
4. $SNO_{2(s)} + 2H_{2(g)} \rightleftharpoons 2H_2O_{(g)}$
   0.4 0.6
   $$K_P = \frac{(PH_2O)^2}{(PH_2)^2} = \frac{0.6 \times 0.6}{0.4 \times 0.4}$$
   $$K_P = \frac{6 \times 6}{4 \times 4} = 2.25$$

9. $H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^- \rightarrow k_1$
   $H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-} \rightarrow k_2$
   $HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{-3} \rightarrow k_3$
   $\underline{H_3PO_4 \rightleftharpoons 3H^+ + PO_4^{-3} \rightarrow k_4}$
   $k_4 = k_1 \times k_2 \times k_3$
   $= 7.10 \times 10^{-4} \times 1.68 \times 10^{-5} \times 4.11 \times 10^{-7}$
   $= 49.02 \times 10^{-16} = 4.902 \times 10^{-15}$
10. $\frac{3}{2}O_{2(g)} \rightleftharpoons O_{3(g)}; k_1 = 2.5 \times 10^{-29}$

    $3O_2 \rightleftharpoons 2O_3 \rightarrow k_2$ $2O_3 \rightleftharpoons 3O_2$
    $\left(\frac{3}{2}O_2 \rightleftharpoons O_3\right) \times 2$
    $3O_2 \rightleftharpoons 2O_3$
    $k_2 = (k_1)^2$

$= (2.5 \times 10^{-29})^2 = 6.25 \times 10^{-58}$

Reverse of k_1

$\left(O_3 \rightleftharpoons \frac{3}{2}O_2\right) \times 2$

$2O_3 \rightleftharpoons 3O_2 \rightarrow k_2$

$k_2 = \frac{1}{(k_1)^2}$

$= \frac{1}{(2.5 \times 10^{-29})^2}$

$= \frac{1}{6.25 \times 10^{-58}}$

$= 1.6 \times 10^{57}$

11. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}; K_{P_1} = 41$

i) $2NH_3 \rightleftharpoons N_2 + 3H_2; K_{P_2} = ?$

$K_{P_2} = \frac{1}{K_{P_1}} = \frac{1}{41} = 2.44 \times 10^{-2}$

ii) $\frac{1}{2}\ N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3; K_{P_3} = ?$

$K_{P_3} = \frac{\sqrt{K_P}}{(K_{P_1})^{\frac{1}{2}}} = (41)^{\frac{1}{2}} = 6.4$

iii) $2\ N_2 + 6H_2 \rightleftharpoons 4NH_3; K_{P_4} = ?$

$K_{P_4} = (K_{P_1})^2 = (41)^2 = 1681$

21. $N_2 + 3H_2 \rightleftharpoons 2NH_3\ \ K_C$

$\frac{1}{2}\ N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3\ K_C^1$

$K_C^1 = \frac{(K_C)^{\frac{1}{2}}}{\sqrt{K_C}}$

Worksheet – 5

Level – I

1. B	2. D	3. D	4. A	5. A	6. B	
7. C	8. A	9. C				

Level – II

1. B	2. A	3. D	4. C	5. A	6. C	7. C	8. C
9. B	10. A	11. C	12. D	13. B,D	14. A,B,C,D		15. B
	16. D						
17. A	18. a-p; b-q; c-r; d-s			19. C	20. A	21. B	22. A
23. B	24. A,B	25. a-p,s; b- p,s; c-q,r; d- q,r					